SIEMENS

Industrial Edge

SIMATIC S7+ Connector

Manual

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Legal information

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Preface

1

1.1 Introduction

1.1.1 Introduction to SIMATIC S7+ Connector

Welcome to SIMATIC S7+ Connector documentation for Industrial Edge.

The SIMATIC S7+ Connector is a south bound connector that connects data from S7-1200/1500 PLCs, PLCSim Advanced and Siemens Software Controller with Industrial Edge. Connectors are essential to collect data from various PLCs and devices and make it available on the Industrial Edge system. With the S7+ Connector data can be accessed via the optimized S7 protocol.

This connector is developed as Connectivity Suite Connector and therefore supports standardized configuration as well as data transfer.

The Common Configurator configures the Connector and is available free of charge in the Industrial Edge Marketplace.

Funding Acknowledgment

Funded by the Federal Ministry for Economic Affairs and Energy based on a decision by the German Bundestag. Your funding provider: Kicks for Edge as part of the EU funding program IPCEI-CIS (FKZ:13IPC008)

1.1.2 Validity of the documentation

The **SIMATIC S7+ Connector for Industrial Edge** documentation is valid for installing the app on an Industrial Edge Device (IE Device). It provides users with comprehensive guidance on how to use the application.

1.1.3 Overview of additional documentation

The following reference links give more information on Industrial Edge applications:

Reference	Description
Industrial Edge Marketplace	Platform to purchase app licenses
Industrial Edge Hub	This page describes the functions of the Siemens Industrial Edge platform and the functionalities of the Edge Management System.All the documentation for the IE platform can be found under **Documents**.
System overview	This page provides an overview of all Edge solutions.
Industrial Edge trainings	Web Based Training: Integrate IT in the production with Siemens Industrial Edge
Databus	Databus documentation
Industrial Information Hub (IIH)	IIH manual
SIMATIC SCADA Export for TIA Portal	More information and the download for the SIMATIC SCADA export.
SIMATIC S7-1500 S7-PLCSIM Advanced	Manual for S7-PLCSIM Advanced

1.2 Legal Information

1.2.1 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit https://www.siemens.com/cybersecurity-industry.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

1.2.2 Security Information for Industrial Edge App

Security information (assumption/constraint) for Industrial Edge Apps are as follows:

- Only authorized internal operators will have access to IE Device with-in secure network using VPN connection.
- Perimeter firewall configuration responsibility lies with end customer.
- Security guidelines for usage of USB sticks within shop floor are applied.
- Creating users with appropriate access rights needs to be done during commissioning and it is the responsibility of the operator.
- Customer is responsible for configuring the application as per the installation/user manual, based on system requirements and technical capabilities of app documented so that the Automation System performance is not impacted.
- The system is installed in an environment that ensures physical access is limited to authorized maintenance personnel only. Managing unauthorized attachment of removable devices is the responsibility of the operator.
- The platform including hardware, firmware and operating system is securely configured and maintained by the operator.
- The operator is capable of protecting the environment from malware infection.
- Centralized IT security components (Active Directory, Centralized IT Logging Server) are provided and well secured by the operator and can be trusted.
- The operator personnel accessing the system is well trained in the usage of the system and general information security aspects like password handling, removable media, etc. are in place.
- Operator is responsible for the Confidentiality, Integrity, and Availability (CIA) of data stored outside the IE Device.
- Operator is responsible for configuring the PLCs with appropriate read/write access levels (Legitimization) and configure Industrial Edge Apps with appropriate passwords for data collection from PLC's.
- Customer takes care about time sync of Industrial Edge Management (IEM) and IE Device.

1.2.3 General Data Protection Regulation (GDPR)

Siemens adheres to the principles of data protection, in particular the principles of data minimization (Privacy by Design).

For this product, SIMATIC S7+ Connector, this means:

1.2.3.1 Personal data

There is no personal data collected but following data is stored to allow machine to machine communication:

- Databus credentials
- OPC-UA server login credentials
- OPC-UA server certificates for encryption and signing
- Tags data and metadata from field devices
- Timestamp
- Smart device information and app usage data

If the customer links the data mentioned above to other data (e.g., shift plans) or if the customer saves personal information on the same medium (e.g. hard disk) and thus creates a personal reference, the customer has to ensure that the guidelines regarding data protection are observed.

> **NOTICE**
>
> This section refers to any personal data processed by the Application other than the personal data contained in log-files / tracking data (if any). **Personal data** are any information relating to an identified or identifiable natural person. Please note that IP-addresses, device identifiers such as IMEI, UDID, IMSI, MAC-address, MSISDN, location data or machine data (if machine data tracks events triggered by user interaction with the machine) usually qualify as personal data.

1.2.3.2 Purpose

The data mentioned above is required for the following purposes:

- Access protection and security measures
- Message system for traceability and availability
- For app diagnosis

Storage of the data is affected for a suitable purpose and is limited to what is strictly necessary, as the information is indispensable in order to identify the authorized operators.

1.2.3.3 Securing of data

The above data will not be stored anonymously or pseudonymized, as the purpose (identification of the operating personnel) cannot be achieved otherwise.

The following data will be used only within the product and within the Edge eco-system and will not be automatically passed on to third parties or unauthorized persons:

- Databus credentials
- OPC-UA server login credentials
- OPC-UA server certificates

The above data is secured by adequate technical measures, such as storing and encryption of process data in databases.

The tags data and metadata from field devices data will be used only within the product and will not be automatically passed on to third parties or unauthorized persons.

The customer must ensure the access protection as part of the process configuration.

1.2.3.4 Deletion policy

This product does not provide an automatic deletion for the Databus or PLC credentials already provided by the user. In case the user provides a different Databus or PLC credentials, the previous credentials will be overwritten. Since there is no explicit delete option, the user could provide junk Databus or PLC credentials to delete the existing valid credentials.

If the user deletes a connection or tag, then the credentials information and other details would also be deleted.

In short, the collected log data will be automatically deleted once the limit is reached (oldest entries first).

1.2.3.5 Data configuration

The customer can configure the data collected via the product using Common Configurator.

What's new

2

2.1 What's new in SIMATIC S7+ Connector?

2.1.1 Version 2.3

2.1.2 Features

- Reduces logging on the ARM platform to error and debug levels only. Info and warning logs are discarded.
- Publishes additional system limits, such as tag subscription limits, in logs and through the device info API.

2.1.3 Bug fixes

- Fixed reading of arrays of bools.
- Fixed checking of system limits for Read Request

2.1.4 Version 2.2

2.1.4.1 Features

- Support for TIA Portal User Management & Access Control (UMAC) authentication.
- Support for larger PLC Programs: The App allows importing or browsing of larger PLC Programs due to extended app memory (requires IE Device OS 2.3).
- Support for ARM IE Devices.

2.1.4.2 Bug fixes

- Fixed writing of string-arrays.

2.1.5 Version 2.1

2.1.5.1 Features

- Supports additional acquisition mode `Cyclic Continuous`.
- Improved application logging: reduced log entries, JSON format.
- Several migration use cases from "SIMATIC S7 Connector" (supported by Import Converter V3.1 and Common Configurator V2.1 or newer).

2.1.5.2 Bug fixes

- Fixed browsing for string-arrays.
- Fixed browsing for negative array bounds.

2.1.6 Version 2.0

2.1.6.1 Features

- The connector can be configured remotely from a central Common Configurator running on IEM System.
- The connector supports browsing of certificates stored on PLCs.
- Support for writing more than 100 data points.

2.1.6.2 Limitations

- Requires IE Device-OS V2.1 or later.

2.1.7 Version 1.5.0

2.1.7.1 Support for Browsing of PLCs

- The connector can browse PLCs with large amount of tags configured.
- Arrays can be browsed as basic data types and as arrays.
- Browsing of arrays of basic types (incl. `String`) provides "array data points" as well as the single elements of the array.

2.1.8 Version 1.4

2.1.8.1 PLC alarms

- PLC alarms reflect now also acknowledged and deleted status of alarms in industrial edge.
- Support more display types for alarm associated parameters, as provided by the TIA portal.

2.1.8.2 Bugfixes

- Connections w/o data points are now supported.

2.1.9 Version 1.3

2.1.9.1 PLC alarms

- The parts of the PLC alarms that are read out at runtime, such as from **ProgramAlarm**, are filled.

2.1.9.2 Supporting special tags

- SIMATIC S7-1500 CPUs supply various kinds of status information independently of the current configuration or of the TIAP project. The S7+ Connector now supports these special tags.

2.1.10 Version 1.2

2.1.10.1 Support for Arrays

- Both array tags and tags for individual array elements are supported by S7+ Connector.

2.1.10.2 Check of PLC certificates

- The PLC certificates are verified. Invalid certificates cause the connection to be rejected.

2.1.10.3 PLC time stamp configuration

- When subscribing to tags, you can choose whether the PLC time stamp or the time stamp of the S7+ Connector is to be used.

2.1.11 Version 1.1

2.1.11.1 CS Registry Service

- Reduced storage footprint.

2.1.11.2 CS Databus Gateway

- Multiple connectors are supported in parallel (including new configuration file format).
- Performance improvements (faster JSON generation)
- In the metadata, **Sample Rate Information** is supported (e.g. for the Audio Connector).
- Improved application log.
- Reduced storage footprint.
- The information **application name** was added in the IE metadata message.
- Improved IE metadata message.
- Improvement of the (re-)connection behavior.

2.1.11.3 SIMATIC S7+ Connector

- Arrays are supported.
- Tested with 20 PLCs in parallel.
- Improved application log.
- Reduced storage footprint.

Overview

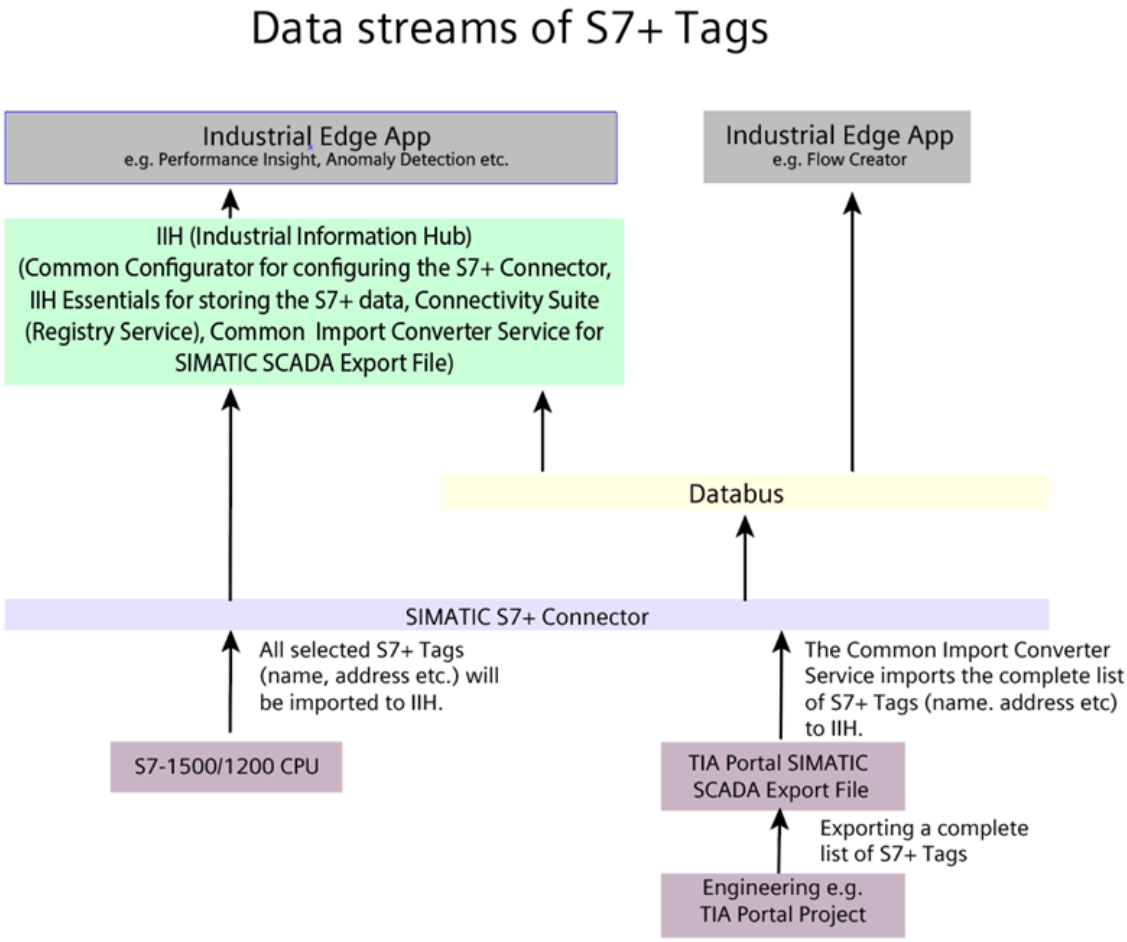
3

3.1 Overview of SIMATIC S7+ Connector

You can use SIMATIC S7+ Connector for Industrial Edge to make the tags of one or more S7-1500/S7-1200 PLCs or from PLCSIM Advanced available in the IIH (Industrial Information Hub) so that the values of the S7+ tags can be analyzed and visualized in apps, such as Performance Insight.

To establish connections, you have to configure them with the Common Configurator. Alternatively you can upload (generated) configuration files via IEM to the connector.

Because SIMATIC S7+ Connector is a Connectivity Suite (CS) connector, you can also make the values of the S7+ tags available directly in the IIH without using the Databus.



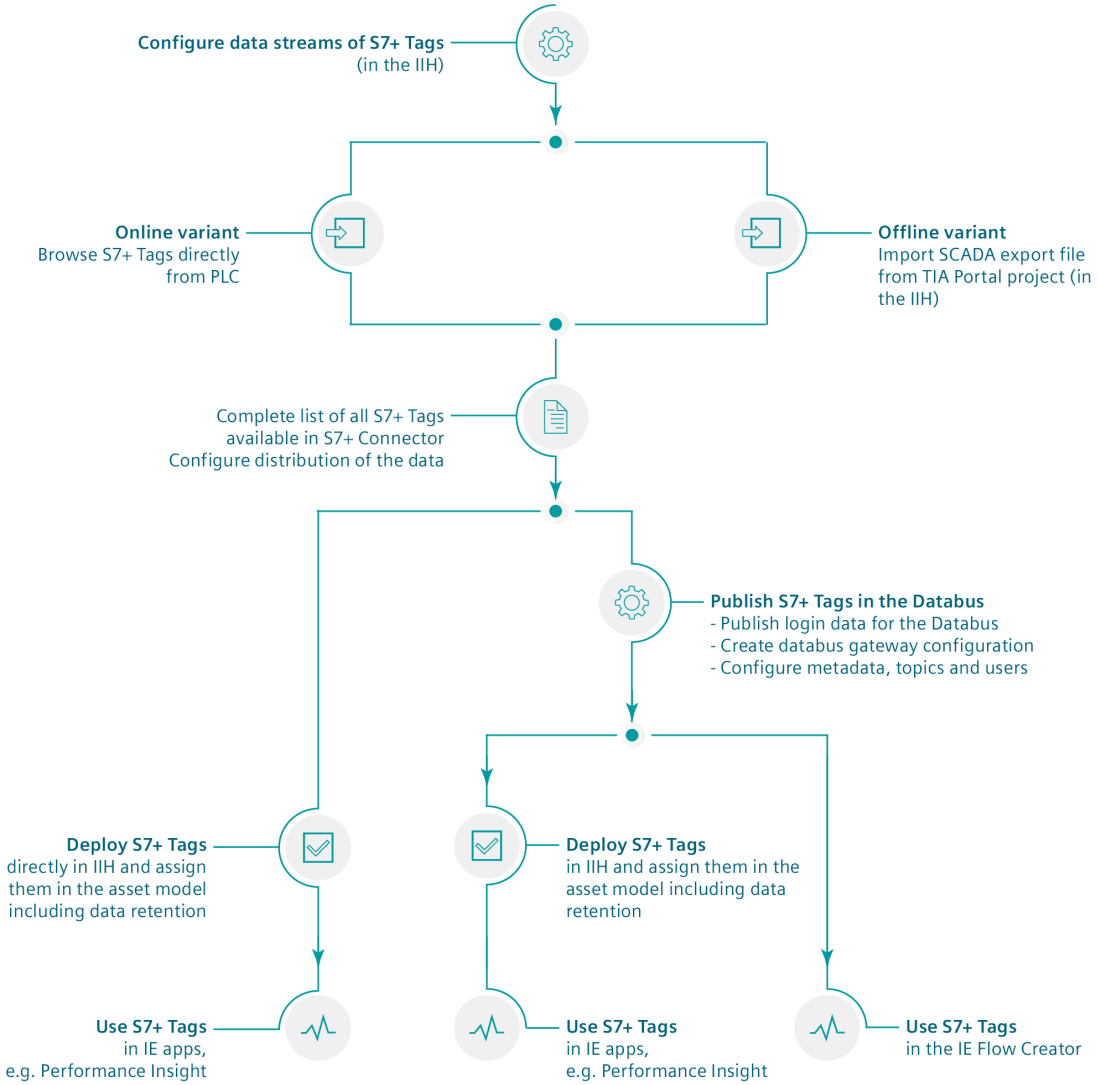
3.1.1 Configuring data streams of the S7+ tags

You can configure the tags for the S7+ Connector using several options:

- Import file from TIAP SCADA Export tool

- Browse tags online
- Import other sources, for example config from IE S7 Connector (Plus CPU)

The possible workflows are shown in the following figure:



3.2 Supported data types

3.2.1 Supported Data Types

Data Types from a PLC must be implicitly converted in order to use them in an IE Device.

The data types are implicitly converted from S7-1500/1200 PLC data types to Connectivity Suite Data Types and then into the corresponding Industrial Edge Data Types:

S7-1500/1200 PLC Data Type	Connectivity Suite Data Type	Length (bits)	Edge Data Type	JSON value Data Type
Bool	Bool	1	Bool	Integer
Byte	UInt8	8	USInt	Integer
Word	UInt16	16	UInt	Integer
DWord	UInt32	32	UDInt	Integer
LWord	UInt64	64	ULInt	String
SInt	Int8	8	SInt	Integer
USInt	UInt8	8	USInt	Integer
Int	Int16	16	Int	Integer
UInt	UInt16	16	UInt	Integer
DInt	Int32	32	DInt	Integer
UDInt	UInt32	32	UDInt	Integer
LInt	Int64	64	LInt	String
ULInt	UInt64	64	ULInt	String
Real	Float32	32	Real	Floating-point number
LReal	Float64	64	LReal	Floating-point number
Char [1)]	String		Char	String
WChar	String		WChar	String
String [1)]	String		String	String
WString	String		WString	String

[1)] SIMATIC PLC store characters for `Char` and `String` with 8 Bits. Remarks see chapter below.

> **Default data types for Industrial Edge**
>
> In the current version of Common Configurator, the Connectivity Suite data types are converted to the corresponding default IE data types.
>
> You can find additional information in the section **Default data types for IE**.

3.2.1.1 Support for Arrays

Arrays are provided in the Common Configurator as a tag.

SIMATIC S7+ Connector supports arrays of default data types, including String, WString and DTL.

S7+ Connector provides all individual elements of an array as separate tags and also the whole array as one common tag in a JSON file. The user can decide which data to work with.

> **Storage of array elements**
>
> There is a limit on the total size of all array elements that can be stored in a PLC. For example, the S7-1516 has a limit of 200 KB. The limit can be different for other models and firmware versions.

3.2.1.2 Support of Char and String

SIMATIC PLC store characters for `Char` and `String` with 8 Bits. The ASCII standard is defined only for 7 Bit. The 2nd half of the possible values is defined by the used code page.

The runtime system has no information about the used code-page in the TIA Portal. That's why the SIMATIC S7+ Connector is using **ISO/IEC 8859-1 (Latin-1)** code-page.

When writing `Char` or `String` values to the PLC this code-page is used as well. Any invalid character is replaced by `U+00FF` (ÿ).

3.2.2 Time data types

3.2.2.1 SIMATIC-specific time data types

SIMATIC-specific time data types are mapped either to `TimeSpan` or to `TimeStamp`:

S7-1500/1200 PLCdata type	Connectivity Suitedata type	Length (bits)	Edge data type	JSON value data type	Commen
S5TIME (s5t) duration Example: S5TIME#10s	TimeSpan Example: **100.000.000**	64	Time Example: **100000000**	Character string (64-bit number as string)	10 [sec] x 1.000 [ms]x 1.000 [us x 10 [100ns]
TIME (t) Example: T#10d20h30m20s630ms	TimeSpan Example: **8.514.263.000.000**	64	Time Example: **8514263000000**	Character string (64-bit number as string)	
LTIME (lt)	TimeSpan	64	LTime	Character string (64-bit number as string)	
Date (d) Example: DATE#2009-12-31	TimeStamp Example: **12.622.176.000.000.000**	64	Date Example: **2009-12-31T00:00:00.0000000Z**	Character string (formatted time string)	7 decimal places because o the [100 ns] resolution
TOD (tod) (TIME_OF_DAY) Example: TOD#10:20:30.400	TimeSpan Example: **372.304.000.000**	64	Time_Of_Day Example: **372304000000**	Character string (64-bit number as string)	
LTOD (ltod) (LTIME_OF_DAY) LTOD#10:20:30.400_365_215	TimeSpan **372.304.003.652**	64	LTime_Of_Day **372304003652**	Character string (64-bit number as string)	1)
DT (dt) (DATE_AND_TIME) Example: DT#2008-10-25-8:12:34.567	TimeStamp Example: **12.249.187.545.670.000**	64	DateTime Example: **2008-10-25T08:12:34.5670000Z**	Character string (formatted time string)	
LDT (ldt) (L_DATE_AND_TIME)	TimeStamp	64	LDT	Character string (formatted time string)	
DTL (dtl)	TimeStamp	64	DateTime	Character string (formatted time string)	

[1] The LTOD data type is mapped to [1 ns] resolution, CS data types: TimeSpan and TimeStamp only in [100 ns], therefore the last two digits are omitted.

> NOTICE
>
> **Default data types for Industrial Edge**
> In the current version of Common Configurator (V1.6), the Connectivity Suite data types are converted to the corresponding default IE data types.
>
> You can find additional information in the section **Default data types for IE**.

3.2.3 Configuring the PLC time stamp

All tags receive a time stamp. You can activate the PLC time stamp (Source Timestamp) or use the connector time stamp for subscribed tags. The connector time stamp is used for non-subscribed tags that are read once.

Both PLC and IE Device must use time synchronization (for example, NTP). Without synchronization, the times of the two devices become out of sync.

3.2.3.1 Requirements

To configure the PLC time stamp, meet the following requirements:

PLC	Minimum firmware version
S7-1500	V2.6 (TIAP V15.1)
S7-1200	V4.4 (TIAP V16)

3.2.3.2 Configuring source timestamp

To configure source timestamp, use the following parameters:

Parameter	Value	Description
`use_source_timestamp`	`True` / `False`	Activates the PLC time stamp.
`max_source_timestamp_diff`	`Int`	The maximum acceptable differential value in seconds. For `0` sec, the difference is not checked. The PLC time stamp is used. For `n` sec, when the difference between the PLC time stamp and the connector time stamp exceeds the set value `n`, the local time stamp is used. When the difference between the PLC time stamp and the connector time stamp is less than the set value `n`, the PLC time stamp is used.

Using the PLC time stamp affects the Quality Code. For more information, see Quality Codes.

NOTICE

When configuring both data subscriptions and alarm subscriptions, it is strongly recommended to enable "Source Timestamp" for your data subscriptions. Alarm timestamps are always generated and provided by the PLC (source timestamp). By using "Source Timestamp" for data subscriptions, this ensures a unified and accurate timeline across all your subscribed information.

3.2.4 Default Data Types in Industrial Edge

In the current version of the Common Configurator (V1.6), the Connectivity Suite data types are published as standard Industrial Edge data types. A different configuration of types is not possible in the current version of Common Configurator.

Connectivity Suite Data Type	Industrial Edge Data Type
Bool	Bool
UInt8	USInt
Int8	SInt
UInt16	UInt
Int16	Int
UInt32	UDInt
Int32	DInt
UInt64	ULInt
Int64	LInt
Float32	REAL
Float64	LReal
TimeSpan LTime (8 bytes)	TimeStamp LDT (8 bytes L_Date_and_Time)
String	String
Raw	Array of Byte

To convert the Connectivity Suite data to the appropriate Industrial Edge data types, you can also configure the Databus Gateway manually. You can find the configuration file in the

Management tab of the Databus Gateway app.

Installing SIMATIC S7+ Connector 4

4.1 System requirements

Note the following system requirements for the installation of the Edge Apps.

4.1.1 Software requirements

The following Internet browsers are required:

- Google Chrome, Version ≥ 72
- Firefox Version ≥ 62

The apps are available in the Industrial Edge Marketplace. From there, you can transfer the latest version of the app to your IEM and then install it on your IE Device:

App/Component	Required version	Description	Source
IE Device-OS	V2.1 or later	Industrial Edge Operating System	IEM System
Common Configurator	V2.0 or later	You need this if, to configure the data sources and data points on IE Device.	Industrial Edge Marketplace
IIH Semantics	V2.0 or later	Optional	Industrial Edge Marketplace
IIH Registry Service	V2.0.0 or later	It creates the connection between the individual components and helps to view the connectors in the Common Configurator.	Industrial Edge Marketplace
Common Import Converter	V3.0.0 or later	It is required for the conversion of different file formats to the IIH compatible format and to import the values of tags from a TIA Portal project offline.	Industrial Edge Marketplace
SIMATIC S7+ Connector	V2.1.0	Mandatory	Industrial Edge Marketplace
Databus	V3.0.0 or later	Required if you want to publish the values of tags to the Databus.	Industrial Edge Marketplace
SIMATIC SCADA Export for TIA Portal	TIA Portal V18 or later	Required if you want to import the values of tags from a TIA Portal project offline.	Siemens Industry Online Support (SIOS)

NOTICE

- All the above mentioned apps must be installed on the same IE Device.
- The latest version of Databus V3.0.0 is capable enough to communicate to connectivity suite connectors as the Databus Gateway application is now integrated to Databus.

4.1.2 Hardware requirements

- A device on which IEM is running.
- An IE Device (IE Device) that is compatible with IEM:
  - IE Device Model: E.g. SIMATIC IPC127E, IPC227E Nanobox, IPC427E, and IPC847E
  - Hard disk: At least 10 GB available
  - RAM: 2 GB RAM available
- The IE Device must be on board the IEM.

NOTICE

IEM, IE Device, and web browsers must be synchronous in the UTC time zone.

4.1.3 Update Databus app

Follow the steps below to update Databus app to V3.0.0:

1. Navigate to the Common Configurator and create a backup of your current configuration.
2. Upgrade the **Databus** app to version 3.0.0.
3. As a precaution, Wait 3 minutes before continuing to ensure all processes are stable.
4. Uninstall the **Databus Gateway** app from your IE Device.

4.2 Check of PLC certificates

4.2.1 Introduction

Certificates ensure secure communication in the system and typically require no user intervention. The TIAP Export includes necessary certificates, while the Common Import Converter embeds them within the JSON configuration. The Common Configurator forwards this file to the S7+ Connector, which validates self-signed and CA certificates.

In cases where certificate changes or deletions are needed, this documentation provides guidance on modifying certificates, adjusting TLS settings, and resolving common certificate-related errors.

NOTICE

SIMATIC S7+ Connector Version 1.1

The secure TLS connection between S7+ Connector and PLC is supported but the PLC certificates are not checked.

SIMATIC S7+ Connector Version 1.2

The PLC certificates are verified.

- If they are not valid, the connection is rejected.
- If no certificates are configured, the check is not performed. The connection is still established and secured as in version 1.1. The communication partners are not authenticated.

4.2.2 Description

You typically do not need to manage the certificates manually. In the certificate management workflow, various components work together to handle and verify certificates automatically:

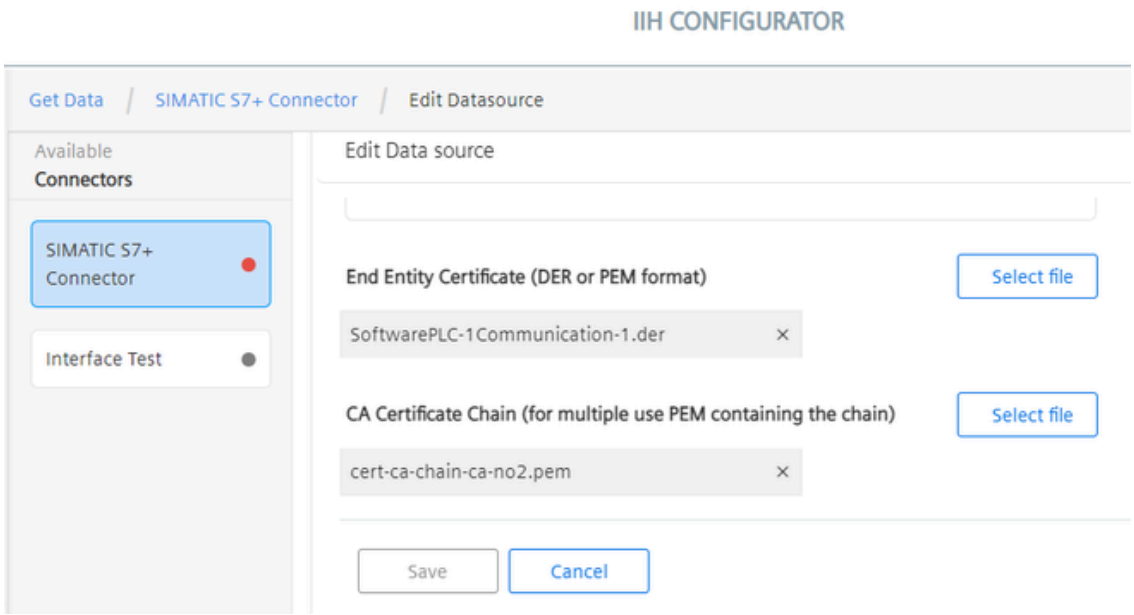
- TIA Portal Export already contains the certificates.

- S7+ Import Converter adds the certificates as an embedded file to the JSON configuration file.
- Common Configurator forwards the JSON configuration file to S7+ Connector.
- S7+ Connector processes the certificates.
  S7+ Connector checks both the self-signed certificates and the CA certificates.

4.2.3 Changing or deleting a certificate

Occasionally, you need to change or delete certificates.

To delete a certificate or provide a different certificate file, use the Common Configurator under 'Datasource Settings'.



NOTICE

When using CA certificates, it is sufficient to provide only the CA certificate chain; the end entity certificate is not needed. Therefore, you can manually delete the end entity certificate in the Common Configurator.

4.2.3.1 Scenarios with CA Certificates

4.2.3.1.1 Ensuring communication with the desired PLC

To ensure that the connector communicates with exactly the desired PLC, both the end entity certificate and the CA certificate chain must be configured. This configuration ensures that communication is specific and secure.

4.2.3.1.2 Using a configuration for multiple Connectors/PLCs

If you want to use a configuration for multiple PLCs, the end entity certificate must be deleted. In this scenario, the driver checks the PLC's certificate against the provided CA certificate chain. This allows for a flexible and scalable configuration option without the need for individual end entity certificates for each PLC.

4.2.4 Possible TLS configurations, changes and errors

TIAP configuration settings	Standard procedure Common Configurator > Import	Possible modifications	Possible causes of an error
TIAP project is not protected. Certificates are not configured.	The JSON configuration file contains the end entity certificate.	The end entity certificate is removed from the JSON configuration file. (S7+ Connector does not check the PLC certificates.)	• The PLC certificate in the JSON configuration file is incorrect. • The PLC certificate in the JSON configuration file has errors.
Self-signed certificate is configured.	The JSON configuration file contains the end entity certificate.	The end entity certificate is removed from the JSON configuration file. (S7+ Connector does not check the PLC certificates.)	• The PLC certificate in the JSON configuration file is incorrect. • The PLC certificate in the JSON configuration file has errors.
A CA certificate is configured.	JSON configuration file contains both end entity certificate and CA certificate (certificate chain).	The end entity certificate is removed from the JSON configuration file. (S7+ Connector checks CA certificates.)	• The PLC and/or CA certificate in the JSON configuration file is incorrect. • The PLC and/or CA certificate in the JSON configuration file has errors.

4.2.5 Possible certificate-related errors

If certificate-related errors occur, check the log file.

Log file	Possible causes
`ERROR Conn-ID'401' 'Login'Callback 'Handle Result' (generic: **Invalid Argument Value**...`	The PLC and/or CA certificate (certificate chain) in the JSON configuration file has errors.
`ERROR Conn-ID'401' - Mismatched end-entity certificate! ERROR Conn-ID'401' 'Login'Callback 'Handle Result' (generic: **TLS Error**...`	The PLC certificate in the JSON configuration file is incorrect.
`ERROR Conn-ID'401' - Error in certificate check! Conn-ID'401' - openSSL error code for certificate check: '19'ERROR Conn-ID'401' 'Login'Callback 'Handle Result' (generic: **TLS Error**...`	The CA certificate (certificate chain) is missing or wrong.

4.3 Installation process

4.3.1 Overview of the installation process

Industrial Edge apps are available in the Industrial Edge Hub Library. The Industrial Edge Hub (IE Hub for short) is the central repository for all available Industrial Edge apps (IE apps) from Siemens and other app partners in the ecosystem.

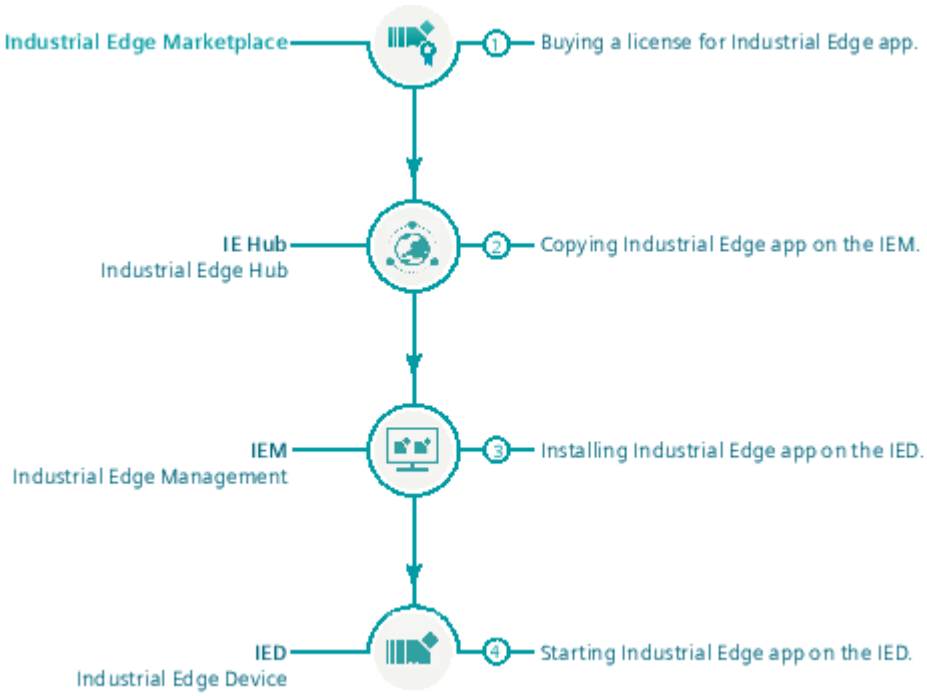
The installation of an IE app involves the following steps:

- Copying an IE app from the IE Hub to IEM
- Installing an app on one or more Industrial IE Devices (IE Devices)
- Starting an installed app on an IE Device

The installation steps are demonstrated in the following video:

4.3.1.1 Overview of the entire IE app purchase process

The following figure shows the steps required to install and start an Industrial Edge app purchased from the Industrial Edge Marketplace on an IE Device:



4.3.1.2 Purchasing a license for an IE app

You can purchase licenses for individual IE apps in the Industrial Edge Marketplace.

You can find detailed instructions at:

How to buy on the Siemens Industrial Edge Marketplace?

You can find more information on purchasing IE apps and managing your account at the following link:

Frequently Asked Questions

4.3.2 Copying an IE app from the IE Hub to the IEM catalog

4.3.2.1 Introduction

This section provides information on how to transfer an IE app from the IE Hub to the IEM catalog of one or more IEM instances.

IEM is a control level for managing all devices, apps, and users of a shop floor.

4.3.2.2 Requirement

The following requirements must be met:

- Internet connection
- Access to the IE Hub
- Available IEM instance

4.3.2.3 Procedure

To copy an app to the IEM catalog, follow these steps:

1. Open the IE Hub and enter your credentials.
   The home screen of the IE Hub is displayed.
2. Open the **Library** tab in the Industrial Edge Hub.
   The library contains all the apps for which you have purchased a license and all system apps.
3. Click the desired app.
   The app details are displayed.
4. In the drop-down list, select all IEM instances to which you want to copy the app.
5. Click **Copy latest version to IEM**.
   The app is copied, and a corresponding job is created. You can follow the job status in the status window of the corresponding IEM instance.

4.3.3 Installing an IE app on the IE Device

4.3.3.1 Introduction

This section provides information on how to install an IE app from an IEM instance on an IE Device.

4.3.3.2 Requirement

The following requirement must be met:

- Access to the IEM instance in whose catalog the IE app to be installed is available.

4.3.3.3 Procedure

To install the IE app on the IE Device, follow these steps:

1. Open the IEM home page and log in.
2. Open the IEM catalog and click the app you want to install.
3. Click the **Install** button.
   The **Install App** dialog window is displayed. If you already have a configuration file, for example, a configuration file that you downloaded from another IE Device, you can upload it here.
4. Select one or more IE Devices on which you want to install the app. The devices must be switched on and online.
5. Select one of the following installation options:
   Click **Install Delayed** or **Install Later** to start the installation at a later time.
   Click **Install Now** to install the app immediately.
6. Click the **Job Status** tab to view the installation progress.

After the installation you can check to see which apps were installed on which IE Devices under **My Installed Apps**.

> Installation of other apps
>
> All the apps can be installed in the same way as explained above.

4.3.4 Starting an IE app on the IE Device

4.3.4.1 Introduction

This section provides information on how to start an IE app installed on the IE Device.

4.3.4.2 Requirement

The following requirement must be met:

- Access to the IE Device on which the IE app is installed.

4.3.4.3 Procedure

To start an app, follow these steps:

1. Open the IE Device home page and log in.
2. Open the **Apps** tab.
3. Launch Common Configurator app and open the Get Data tab.
   The **SIMATIC S7+ Connector** app appears in the available connectors list.
4. Click **SIMATIC S7+ Connector** app to start configuring the data sources and tags.

Configuring SIMATIC S7+ Connector

5

5.1 Connector Configuration

5.1.1 Configuring SIMATIC S7+ Connector

5.1.1.1 Overview

You can configure the SIMATIC S7+ Connector in two ways:

- Online configuration
- Offline configuration

The diagram below shows the online and offline configuration options:

```
flowchart TD
  CFG[Configure <br>S7+ Connector] --> OFFL[<b>OFFLINE</b><br> Configuratio
  CFG --> ONL[<b>ONLINE</b><br> Configuration]
  OFFL --> MIGR[<b>Migration</b><br>S7 Connector Config]
  OFFL --> TIAP[<b>Engineering Tool</b><br>TIAP Scada Export]
  ONL --> PLC[<b>Browsing</b><br>Online Access to PLC]
```

5.1.1.2 Involved components

The offline use case involves the following components:

- Common Configurator as UI for configuring
- TIA Portal including TIAP SCADA Export
- Common Import Converter (Background Service as Edge App)

The offline use case also includes migration scenarios. You can migrate existing configurations of the SIMATIC S7 Connector when S7-1200/1500 was configured using the optimized S7 protocol.

The online use case involves the following components:

- Common Configurator as UI for configuring
- S7+ Browsing Service (integrated in SIMATIC S7+ Connector App)

The diagram below shows the involved components for online and offline configuration:

```
flowchart TD
  TIA[TIA Portal] --> IMP[Common Import Converter]
  IMP --> CC[Common Configurator]
  CC --> S7CONN[SIMATIC S7plusConn]
  S7CONN --> CC
  CC --> BRWS[S7+ Browse Service]
  BRWS --> CC
  S7CONN --> PLC[SIMATIC PLC]
  PLC --> S7CONN
  BRWS --> PLC
  PLC --> BRWS
```

5.1.1.2.1 Common Import Converter

This tool converts vendor and device-specific configuration files into standardized Connectivity Suite-compatible configurations. It also converts TIAP export files to Connectivity Suite JSON configurations.

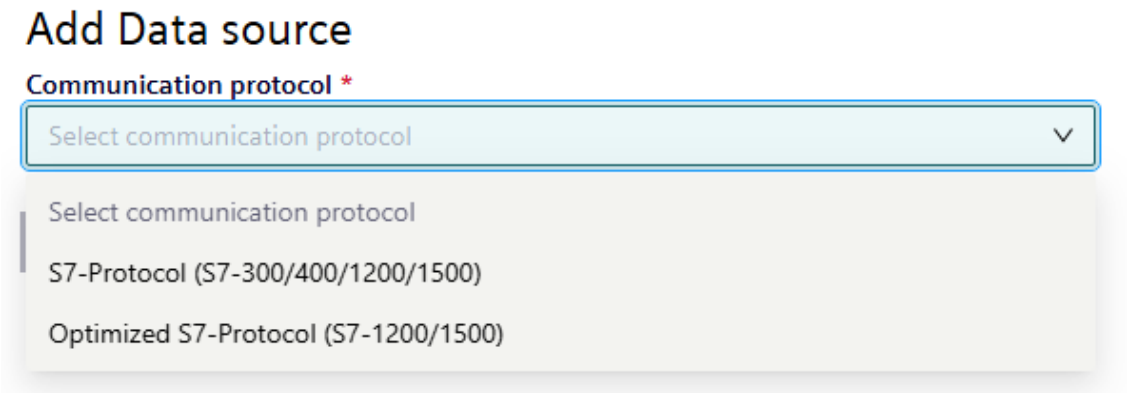
The Common Configurator uses this conversion functionality when you import a device-specific configuration. The Common Configurator runs the Common Import Converter in the background to convert the file into the common JSON configuration format.

5.1.1.2.2 Migrating SIMATIC S7 Connector

The SIMATIC S7 Connector supports the following configuration variants:

- S7-300/400, S7 protocol
- S7-1200/1500, S7 protocol
- S7-1200/1500, optimized S7 protocol

The image below shows the supported protocols of SIMATIC S7 Connector:



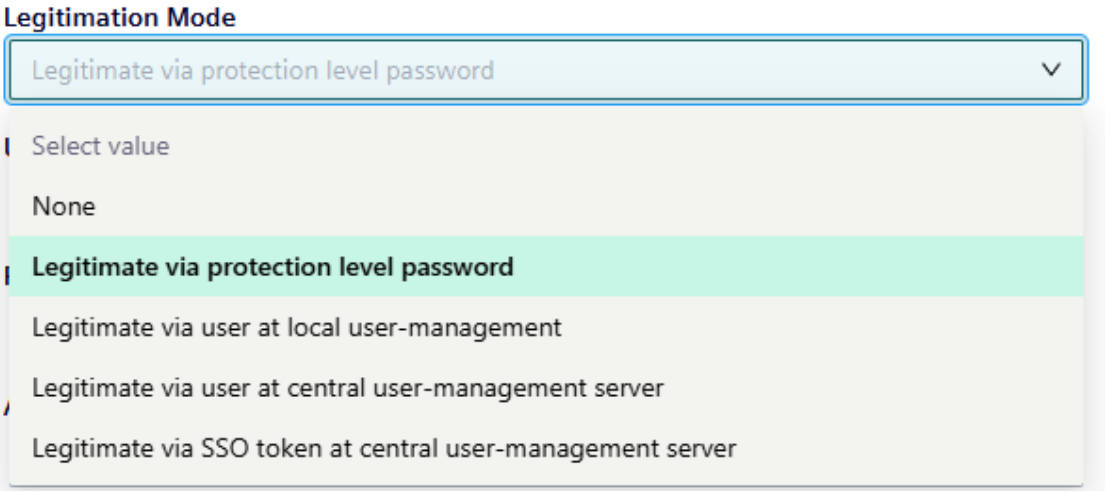
NOTICE

Migration from SIMATIC S7 Connector to SIMATIC S7+ Connector is only supported for the configurations of S7-1200/1500, Optimized S7-Protocol.

5.1.1.2.3 Legitimation modes

The SIMATIC S7+ Connector supports multiple UMAC (User Management & Access Control) legitimation modes:

- Legacy (default): Legitimate via protection level password. The username is ignored.
- Local User: Legitimate via user at local user-management (TIAP V19+, CPU FW 3.1+).
- Central User: Legitimate via user at central user-management server (TIAP V20+, CPU FW 4.0+).
- Token: Legitimate via SSO token at central user-management server. The username is ignored (TIAP V20+, CPU FW 4.0+).



> NOTICE
>
> The legitimation via token with limited lifetime is not recommended, as it needs to be updated after expiration.

For more information, see:

- TIAP UMAC
- Central User Management

5.1.2 SIMATIC S7+ Browse Service

5.1.2.1 Overview

The SIMATIC S7+ Browse Service is not a separate edge application. It is integrated in the SIMATIC S7+ Connector since version V1.5.

You need this service to configure the SIMATIC S7+ Connector when it is online and connected to a PLC. This means you do not need a TIAP project or export for this use case.

The diagram below shows the components and data flow when using S7+ Browsing Service:

```
flowchart TD
    CC[Common Configurator] --> |<i>1. Browse Request</i>| BRWS[SIMATIC S7+
    BRWS --> |<i>2. Online Access PLC</i>| PLC[SIMATIC PLC]
    PLC --> |<i>3. Data points</i>| BRWS
    BRWS --> |<i>4. Browse Response</i>| CC
```

5.1.2.2 Supported functionality

The SIMATIC S7+ Browse Service supports the following:

- S7-1500 CPUs and S7-1200 CPUs from FW V4.0
- All data types including `DTL`
- Arrays and structs by providing all single elements
- Basic-type arrays as array objects
- PLC passwords in connection parameters
- End-entity certificates and CA certificates in connection parameters

5.1.2.3 Workflow and use cases

You can use the browse service in two ways:

- Modifying an existing data source
- Creating a new data source

5.1.2.3.1 Modifying an existing data source

When the data source already exists, all required connection parameters are in place. This includes IP address, PLC password, and certificate files. You can use the browse functionality of the Common Configurator.

5.1.2.3.2 Creating a new data source

You must provide the IP address of the PLC.

When the PLC is configured with no access (complete protection), you need at least HMI access to browse the data points.

Browsing works when you do not configure any certificates.

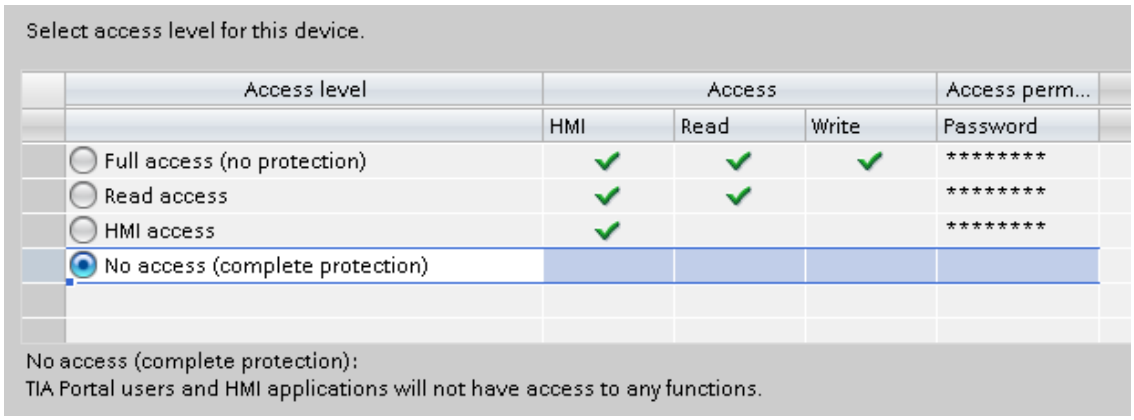
5.1.2.4 Credentials

5.1.2.4.1 General issues

Scenario	API Call Return Code
PLC not reachable	`UNAVAILABLE (14)`
Invalid IP (192.1.1.300)	`INVALID_ARGUMENT (3)`

5.1.2.4.2 CPU password

A CPU password is required only when the PLC is configured with no access (complete protection). You need at least the password for HMI access.



CPU Access Level	Browse Request	Browse Response
Full access	no PW	✓ browsed nodes
Full access	PW configured	✗ Error `PERMISSION_DENIED (7)` Legitimate to OMS+ Session rejected. Check password.
Read access	no PW	✓ browsed nodes
HMI access	no PW	✓ browsed nodes
No access	no PW	✗ Error `PERMISSION_DENIED (7)` Explore PLC Tag Containers failed
No access	HMI password	✓ browsed nodes

NOTICE

You can check the detailed error messages in the application log of the SIMATIC S7+ Connector and SIMATIC S7+ Browse Service.

5.1.2.4.3 Certificates

When establishing an online connection to the PLC, the browsing service considers certificates only when they are part of the connection parameters.

> **NOTICE**
>
> Both End-Entity Certificates and CA Certificates are supported.

CPU Certs	Certs Browse Request	Browsing Result
EE Cert	no cert	✓ browsed nodes Auto-Trust WARNING: Accept unknown certificate due to missing configuration
EE Cert	valid EE cert	✓ browsed nodes
EE Cert	invalid EE cert	❌ Error `INVALID_ARGUMENT (3)` Error Invalid Argument Value
EE Cert	wrong EE cert	❌ Error `UNAUTHENTICATED (16)` Error TLSVerificationFailed
CA Cert(s)	no cert	✓ browsed nodes Auto-Trust WARNING: Accept unknown certificate due to missing configuration
CA Cert(s)	valid CA cert(s)	✓ browsed nodes
CA Cert(s)	invalid CA Cert(s)	❌ Error `INVALID_ARGUMENT (3)` Error Invalid Argument Value
CA Cert(s)	wrong CA cert(s)	❌ Error `UNAUTHENTICATED (16)` Error TLSVerificationFailed

> **NOTICE**
>
> The browsing service does not return PLC certificates as part of the browsed data (`Browse-API`). When an application needs the certificate information from the PLC, it can use the `DeviceInfo-API` from the Browse Service (as used by the Common Configurator).

5.1.2.5 Limitations

The browse service has the following limitations:

- Certificates are not provided in the browsing information.
- Alarm classes are not provided.

> **NOTICE**
>
> When something goes wrong, you must check the application log of the **SIMATIC S7+ Connector** first, followed by the **SIMATIC S7+ Browsing Service**. The logs of the browsing are part of the S7+ Connector logs.

5.1.3 Common Import Converter for SIMATIC S7+ Connector

The Common Import Converter is installed as a background service. You do not work directly with this app. The Common Configurator calls it in the background. The converter is used when a configuration file needs conversion to the format used by the Common Configurator.

5.1.3.1 Overview

The Common Import Converter offers a wide range of functionality for importing or migrating use cases. This chapter covers the following use cases related to SIMATIC S7+ Connector:

- Importing TIA Portal SCADA Export files
- Migrating configurations from S7 Connector to SIMATIC S7+ Connector (only S7-1200/1500, optimized S7 protocol)

5.1.3.2 Using Import Converter with Common Configurator

The diagram below shows the data flow and involved components when configuring offline:

```
flowchart LR
    USER[<b>User</b><br><i>Restore a Backup / <br>Add Data Source from File
    CC --> |<i>Request <br> Conversion</i>| CNV[Common <br>Import Converter
    CNV --> |<i>Standardized <br>Config JSON</i>| CC
```

5.1.3.3 Supported functionality

The Common Import Converter supports the following:

- S7-1500 CPUs and S7-1200 CPUs
- All data types including `DTL`
- Arrays and structs by providing all single elements
- Basic-type arrays as array objects
- PLC passwords
- End-entity certificates and CA certificates

5.1.3.4 Using TIA Portal SCADA export

```
flowchart LR
    TIAP[TIA Portal <br> incl. <br> TIAP SCADA Export] --> |<i>Export <br>c
    ZIP --> |<i>Provide ZIP</i>| CC[Common <br>Configurator]
    CC --> |<i>Request <br> Conversion</i>| CNV[Common <br>Import Converter
    CNV --> |<i>Standardized <br>Config JSON</i>| CC
```

5.1.3.4.1 Workflow with TIA Portal SCADA export

To use TIA Portal SCADA export, proceed as follows:

1. Export the PLC from TIA Portal with SIMATIC SCADA Export for TIA Portal. For more information, see SIOS.
   - The `ZIP` file contains several encrypted `JSON` files.
   - The Common Import Converter processes the `ZIP` file and provides a common JSON configuration file.
2. Use the Common Configurator to import (upload) the `ZIP` file.
3. The Common Configurator runs the Common Import Converter in the background.

> **NOTICE**
>
> When something goes wrong, you must check the application log of the **Common Import Converter**. You can download the application log via IE Device or via IEM.

For more information, see Configuring with the Common Configurator.

5.2 Configuring data streams

5.2.1 Configuring with the Common Configurator

The SIMATIC S7+ Connector allows you to configure, manage, and maintain data sources and tags.

For detailed instructions on managing data sources and tags, see Industrial Information Hub user manual.

This section provides resources for managing data sources and tags. It includes instructions on configuring, adding, editing, and deleting data sources and tags. It also includes steps for deploying them. You will also find guidance on importing tag configurations and backing up or restoring them.

The Common Configurator covers the following use cases related to the SIMATIC S7+ Connector:

- Configuring data sources offline (TIA Portal Project export)
- Configuring data sources online (online access to PLC required)
- Migrating from existing SIMATIC S7 Connector configuration files

5.2.1.1 Importing tags offline

To define and configure tags for connected PLCs in the Common Configurator, export the tags from a TIA Portal project and import them into the Common Configurator.

All available tags are then displayed in the Common Configurator.

Loading the tags may take some time depending on the size of the export file.

5.2.1.1.1 Requirements for importing TIAP configuration

Before importing TIAP configuration, meet the following requirements:

- Common Import Converter is installed on the IE Device
- A TIA Portal project with configured PLCs and tags is available
- SIMATIC SCADA Export for TIA Portal is installed
  This TIA Portal Add-On exports all required information from the TIA Portal project and generates an export file for the Common Configurator. For more information and download, see SIMATIC SCADA Export for TIA Portal.
- The connections to the desired PLCs have been configured in the Common Configurator.

> NOTICE
>
> If a PLC has more than one Ethernet interface, the import includes all IP addresses. Before proceeding, delete the addresses you do not need.

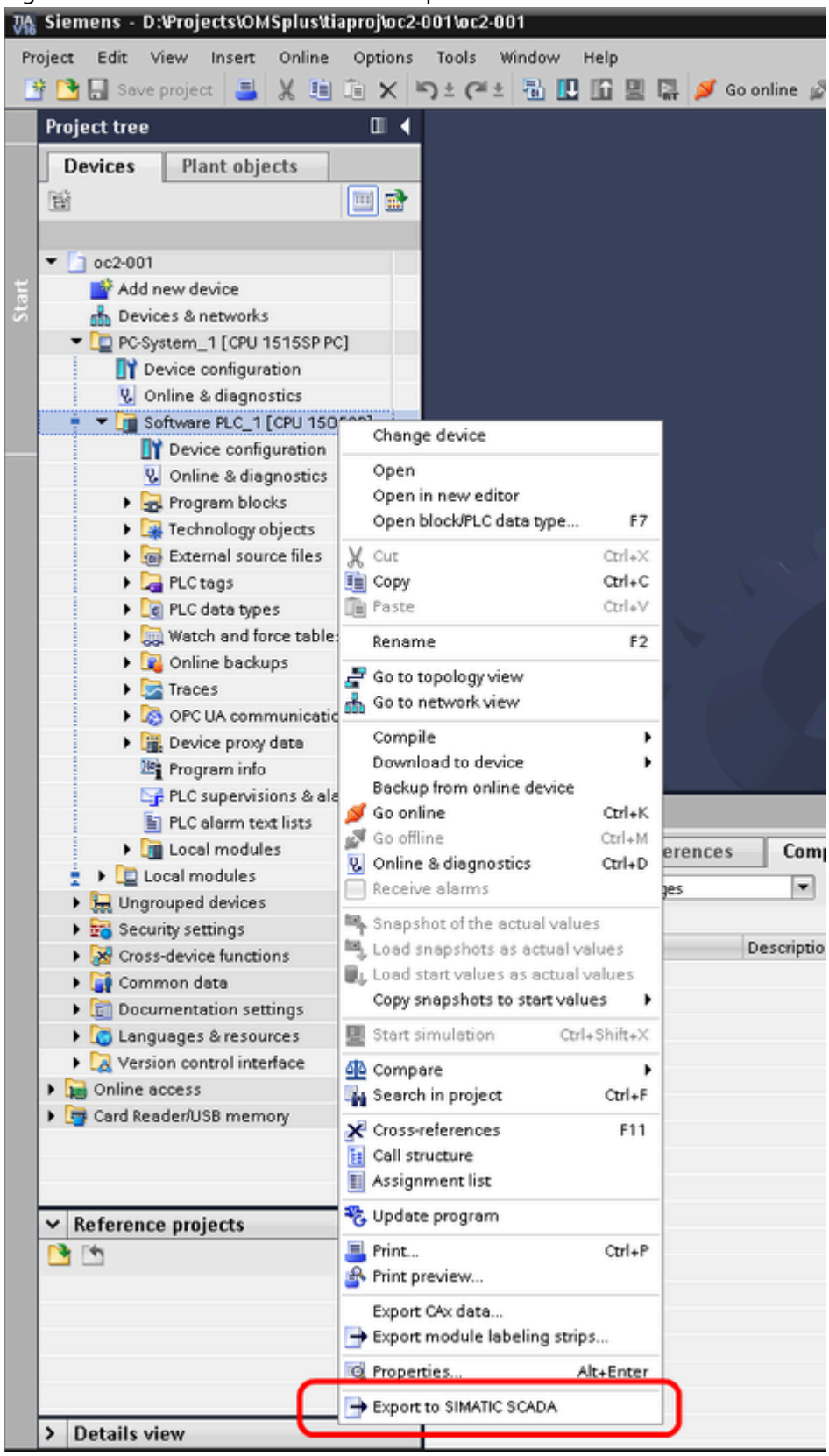
5.2.1.1.2 Exporting tags from the TIA Portal project

To export the tags from the TIA Portal project, proceed as follows:

1. Download the Add-On Tool SIMATIC SCADA Export for TIA Portal from Siemens Industry Online Support and install it on the PC where the TIA Portal runs.
2. Compile the TIAP project.

> **Compiling TIAP project**
>
> Compile the TIAP project each time before exporting so that the export provides consistent data.

3. Right-click the desired PLC and select "Export to SIMATIC SCADA" from the shortcut menu.



> **NOTICE**
>
> Export Tags for only one PLC at a time.

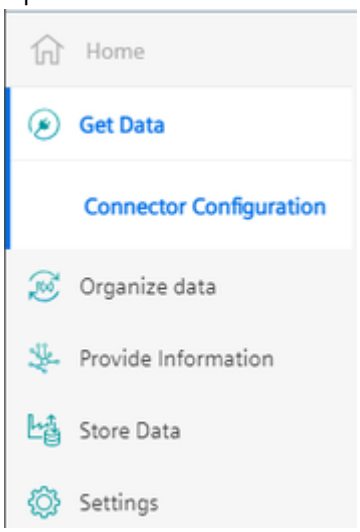
An export file is created.

4. Repeat the process for additional PLCs if necessary.

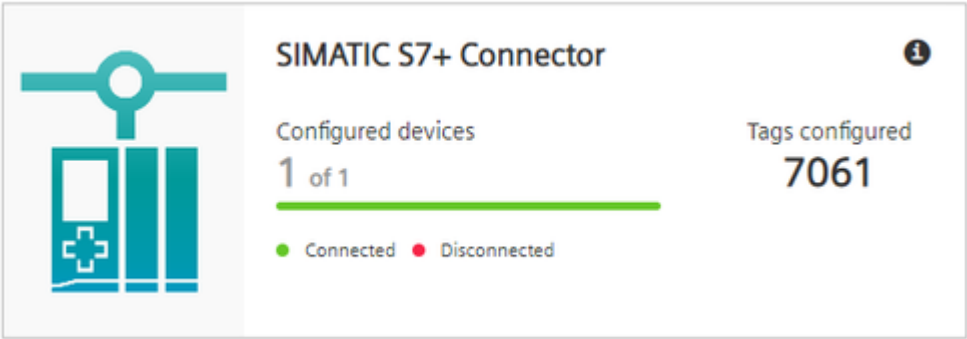
5.2.1.1.3 Importing tags

To import the tags from the SIMATIC SCADA export file, proceed as follows:

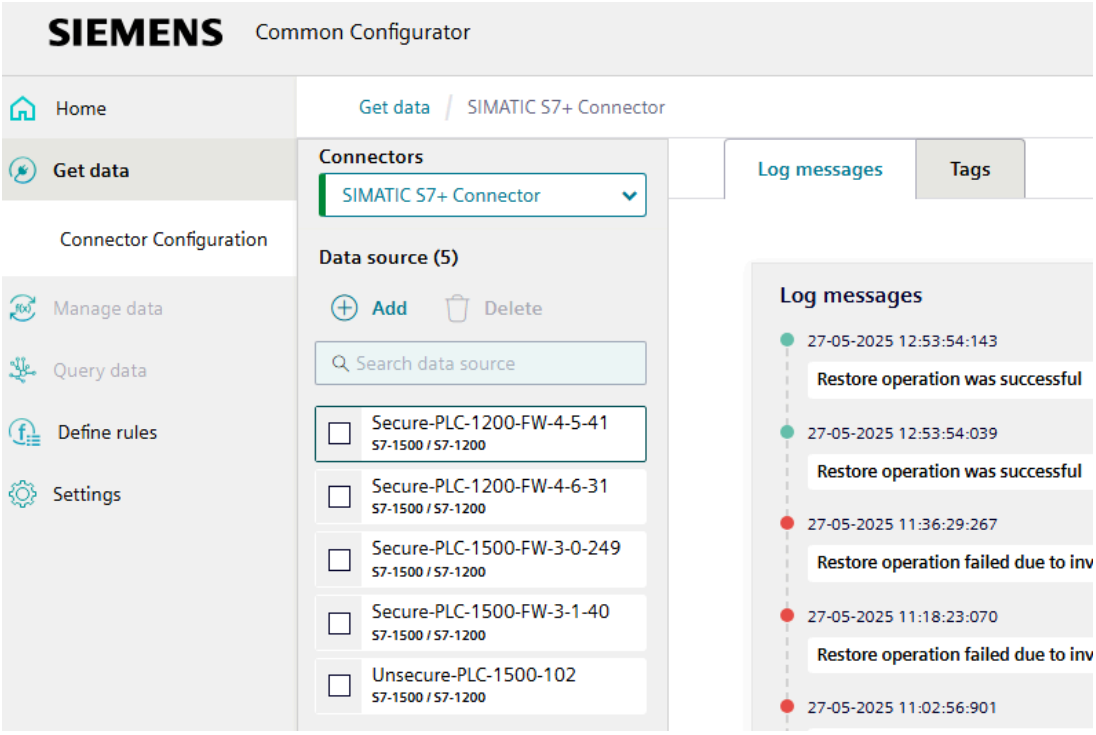
1. Open the "Get Data" tab in the IIH.



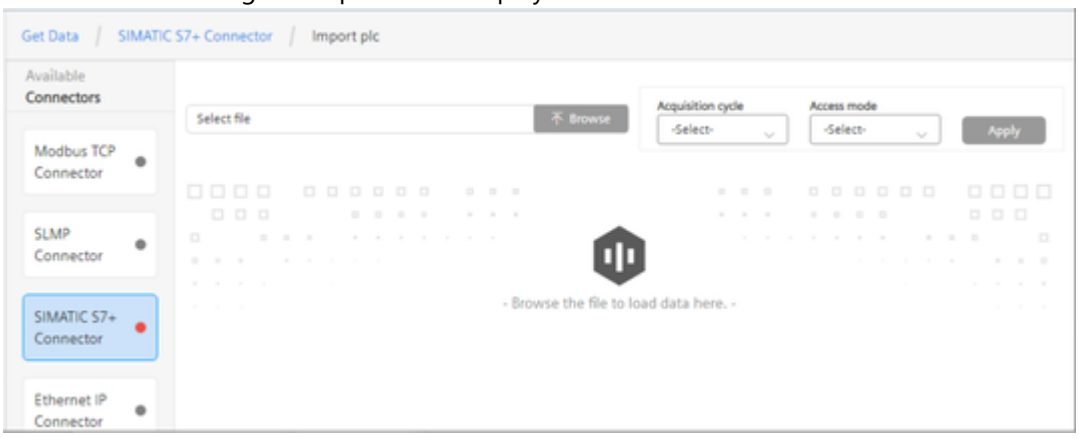
Installed or available connectors appears. A SIMATIC S7+ Connector can access multiple PLCs at the same time. You can see at a glance how many configured PLCs have been connected and how many tags are transferred.



2. Click "SIMATIC S7+ Connector".
   The SIMATIC S7+ Connector page appears.

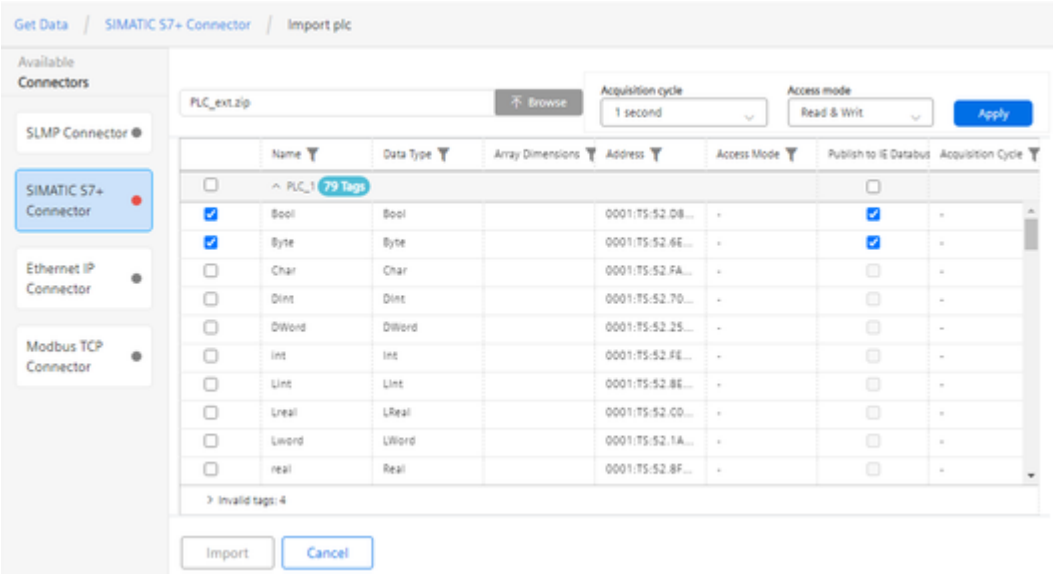


3. Open the "Tags" tab.
4. Click Import.
   The view for selecting the import file is displayed.



5. Click Browse and select the import file.
   This button is only displayed when you have installed the Common Import Converter on the IE Device.

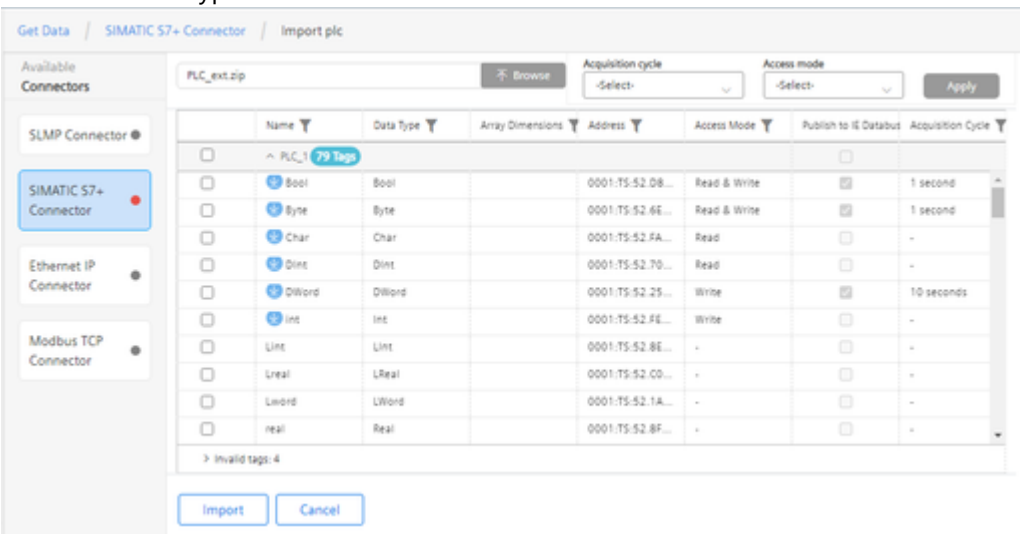
The tags included in the exported project are displayed.



NOTICE

- Tags that can not be imported are listed under **Invalid Tags**.
- If the tags are imported from an unknown data source, the new data source is created.
- If a data source with the same name already exists, the imported tags are added to the tags already existing.

6. Select "Acquisition cycle" and "Access type".
7. Select the tags to publish in the IIH.
8. Click "Apply".
9. Repeat the procedure for further tags to import with a different acquisition cycle or with a different access type.



10. Click "Import".
    The imported tags are displayed in the "Tags" view.

5.2.1.1.4 Result

The imported S7+ tags are displayed.

Then deploy the connection to the PLC and the selected tags to the IIH Semantics. For more information, see Deploying S7+ tags in IIH Semantics.

5.2.1.2 Importing tags online

5.2.1.2.1 Description

When you have an online connection to the PLC, you can browse and import tags directly.

Loading the tags may take some time depending on the size of the project.

5.2.1.2.2 Requirements

You need an online connection to the PLC.

5.2.1.2.3 Importing tags

To import tags online, proceed as follows:

1. Open the "Tags" tab.
2. Right-click the icon to the right of the desired PLC.
3. Click Browse Tags.
4. Select the desired tags.

5.2.1.2.4 Result

The imported S7+ tags are displayed.

Then deploy the connection to the PLC and the selected tags to the IIH Semantics. For more information, see Deploying S7+ tags in IIH Semantics.

5.2.1.3 Configuring the acquisition mode

The S7+ Connector supports the following acquisition modes:

- OnChange
- Cyclic Continuous

> NOTICE
>
> This setting is configured at the **connection level**.

5.2.1.3.1 OnChange

The default setting is OnChange. In OnChange mode, data is only transmitted when values change. This configuration is optimal for system load for all involved components such as the network, Databus, IIH Essentials, or IIH Semantics.

The S7+ Connector operates internally on a subscription-based model. This means that the PLC implements change-of-value logic.

5.2.1.3.2 Cyclic Continuous

Some specific use cases might require Cyclic Continuous, such as machine learning applications. The cyclic continuous data acquisition mode sends all data upstream each cycle, even when the data does not change. This mode impacts the performance of the IE Device. The northbound gRPC connections to the Databus and/or IIH Essentials / IIH Semantics experience higher load. Storage utilization for the timeseries data in IIH Essentials also increases. The communication with the PLC is also heightened.

5.2.1.3.3 Mixed scenarios

When selected data points are needed in Cyclic Continuous mode, define a separate data source with Cyclic Continuous. This way, the common data points can be efficiently provided in the OnChange mode in a separate data source. Meanwhile, only the required data points are provided in Cyclic Continuous mode.

5.2.1.4 Deploying S7+ tags in IIH Semantics

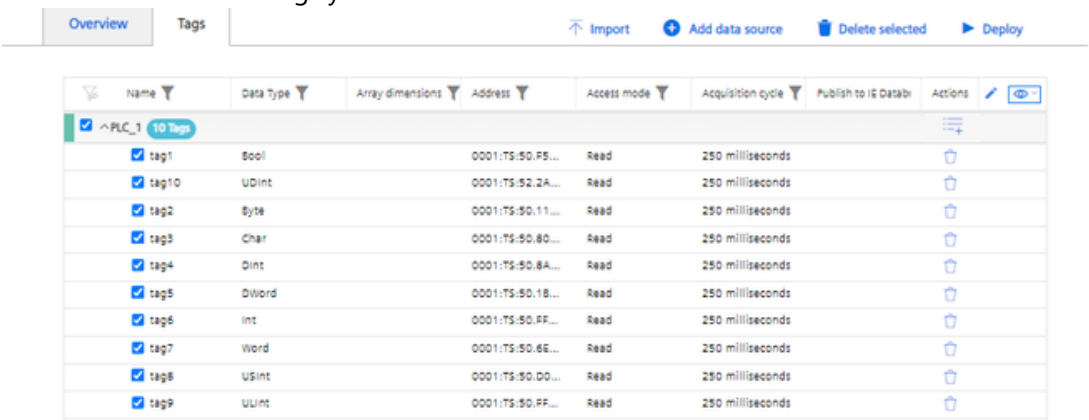
5.2.1.4.1 Description

Once you configured the S7+ tags, activate the tags you need and deploy them in IIH Semantics.

5.2.1.4.2 Deploying tags

To deploy tags, proceed as follows:

1. Select the PLC and the tags you want to activate in the IIH.



2. Click "Deploy".

5.2.1.4.3 Result

The IIH generates the configuration file for the SIMATIC S7+ Connector in the background. The configuration file is sent directly to the connector. You have now defined the tags that are available for the IIH.

You can find the configuration file in the "Management" tab of the IE Device.

5.2.2 Configuring via Databus (optional)

5.2.2.1 Entering the credentials for Databus (Common Configurator)

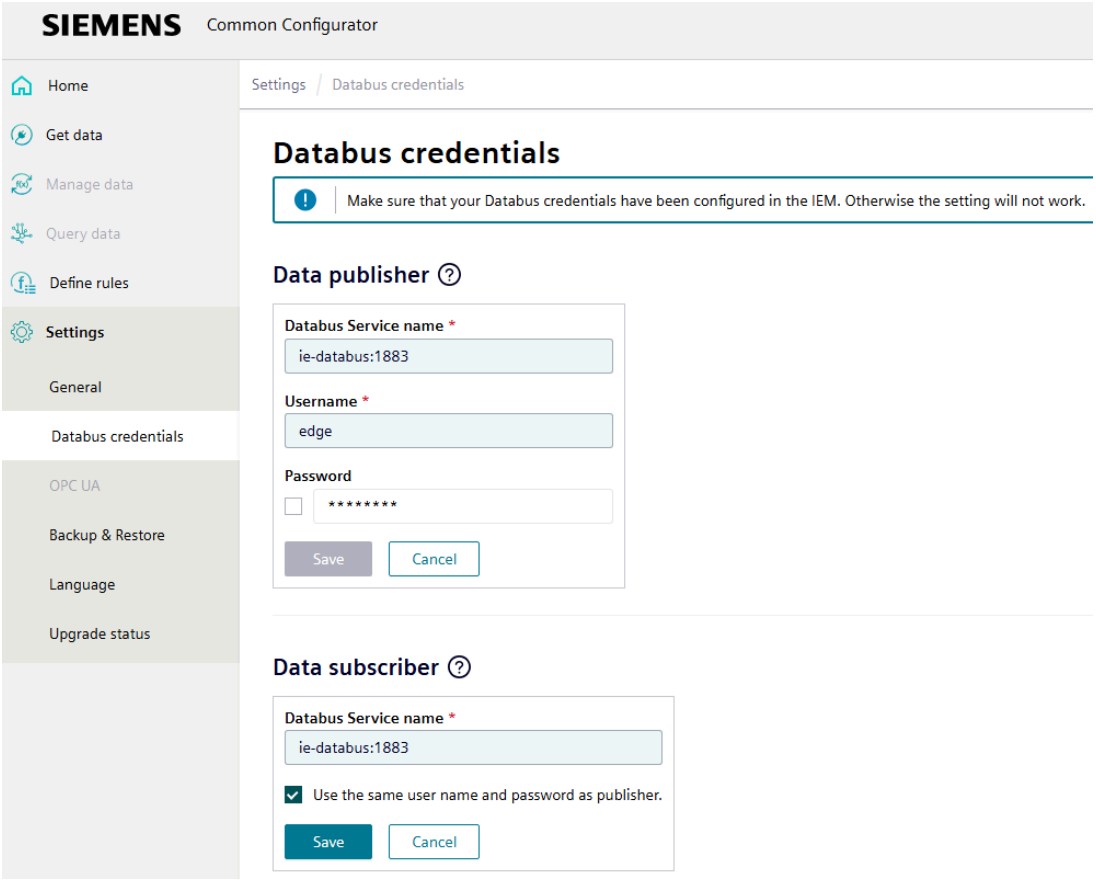
You need to enter the credentials to publish the S7+ data in the Databus.

When you want to make the S7+ data directly available in the data store of the Industrial Information Hub, you do not need to enter any credentials.

5.2.2.1.1 Entering the credentials

To enter the credentials, proceed as follows:

1. Open the "Settings" tab.
2. Enter a Databus Service name.
3. Enter the corresponding credentials in the "Data Publisher Settings" tab.



4. Click "Save".

5.2.2.1.2 Result

The connection to the Databus is established.

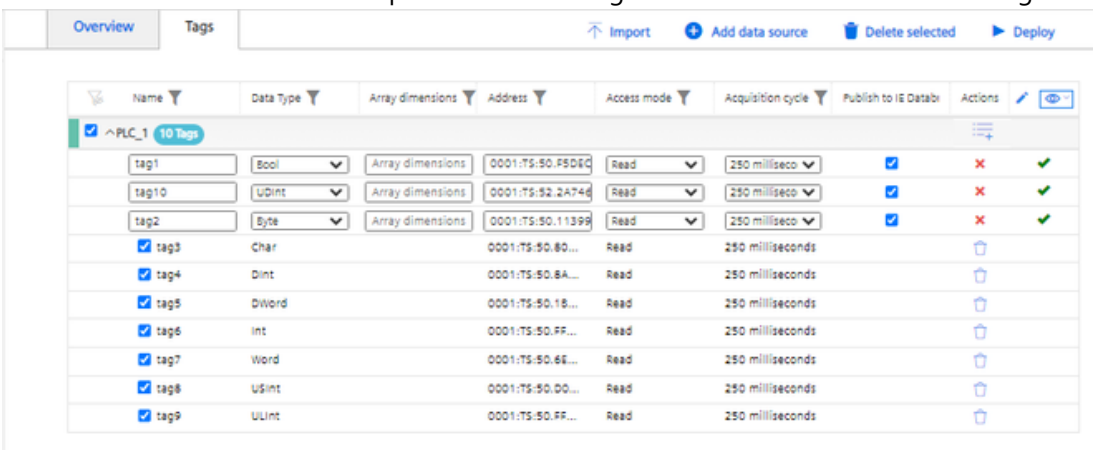
5.2.2.2 Deploying S7+ tags in the Databus (Common Configurator)

S7+ tags become available in the Databus when you enable the corresponding option for selected tags.

5.2.2.2.1 Deploying tags

To deploy tags in the Databus, proceed as follows:

1. Select the "Publish to Databus" option for all S7+ tags or for individual selected S7+ tags.



2. Click "Deploy".
   The configuration file for the Databus Gateway is created automatically.

5.2.2.2.2 Result

The IIH generates the configuration file for the SIMATIC S7+ Connector in the background and sends it to the connector. This defines which tags are available for the Databus Gateway. A connection is established by configuring the Databus Gateway.

The Databus subscribes to the tags that the Databus Gateway found and forwarded to the Databus.

You can find the configuration files in the "Management" tab of the IE Device.

> **Credentials for Databus**
>
> Enter the credentials in the settings to enable the connection to the Databus.

5.2.2.3 Configuring subscriptions manually from the PLC (Databus Gateway)

The Databus Gateway is the bridge between Industrial Edge and the Connectivity Suite. This allows you to make the tags available in the Databus.

For information about automatic generation of the Databus Gateway configuration, see Deploying S7+ tags in the Databus (Common Configurator).

You can generate the configuration yourself (for example, using a code generator). You must transfer the configuration file to the IE Device.

Define which tags to publish on the Databus. Specify the following parameters in the Databus Gateway configuration:

- Which PLCs or connections to use
- Which tags to subscribe (cyclic reading)
- The metadata topic of the Databus on which to publish the tags

5.2.2.3.1 Updating apps

When updating apps, consider the following:

- When you update Common Configurator, also update the Databus Gateway.
- When you only deploy tags directly to the IIH, the Databus Gateway is not required.

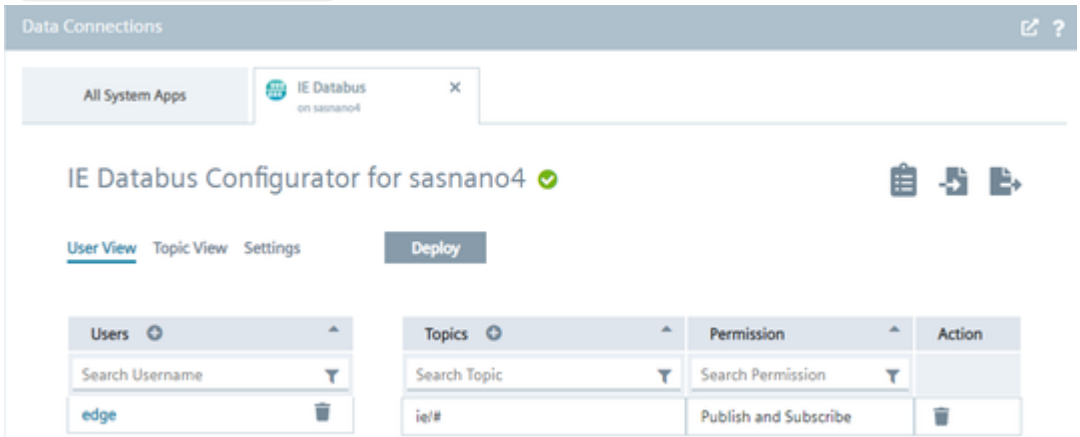
5.2.2.3.2 Requirements

The tags for the PLC have been defined and activated.

5.2.2.3.3 Configuring the connection

To configure the connection to the Databus Gateway, proceed as follows:

1. In the "Tags" tab, click "Publish to Databus".
   The configuration file is automatically created for the Databus Gateway.
2. When installing the Databus Gateway app in the IEM, click "+" to create a configuration file (`cs-connector-config.json`).



3. Specify the following parameters:
   - `name`: Name of the PLC
   - `driver_instance`: Name of the connector instance

- `connection_name`: Must match the name of the data source that was configured in the IIH.
- `data points > name`: Must match the name of the tag that was configured in the IIH below the data source.

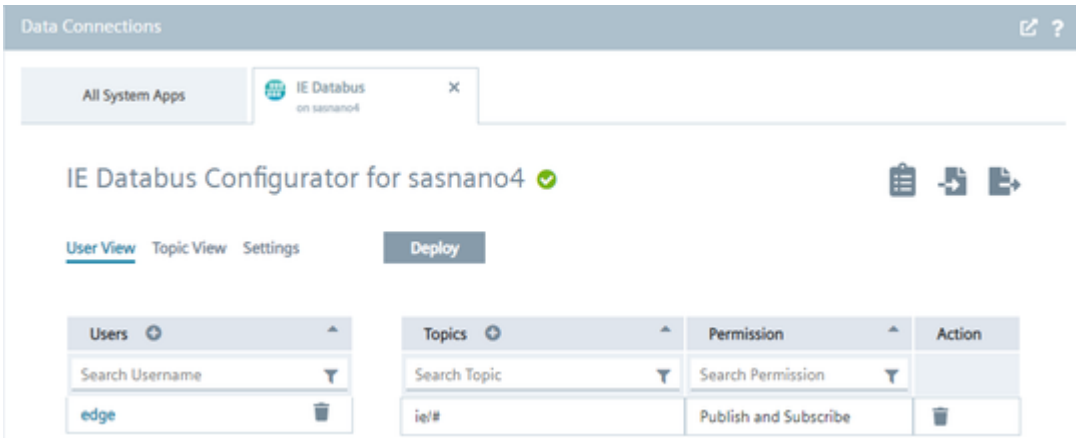
You can find the configuration file in the "Management" tab of the Databus Gateway app.

5.2.2.3.4 Result

You have defined which tags to publish on the Databus.

5.2.2.4 Configuring metadata topics and users (Databus)

To use the data of the SIMATIC S7+ Connector via the Databus, you must also configure the Databus. In the configuration, subscribe to the metadata topics used by the SIMATIC S7+ Connector.



For more information on the Databus configuration, see Databus.

> **User name and password**
>
> Make sure that the user name, password, and metadata topic that you configured in the Databus Gateway exactly matches with these entries in the Databus configuration.

5.2.3 Migrating from SIMATIC S7 Connector

> **NOTICE**
>
> The migration from SIMATIC S7 Connector to SIMATIC S7+ Connector is only supported for S7-1200/1500 controllers that are configured with Optimized S7-Protocol.

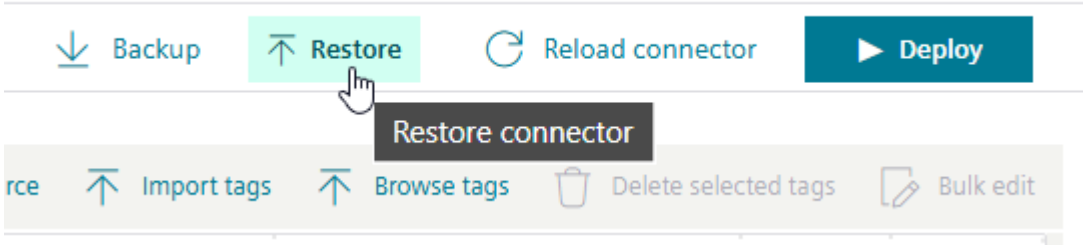
You can migrate an existing configuration from SIMATIC S7 Connector in two ways:

- Migrate at connector level
- Migrate at connection level

5.2.3.1 Migrating at connector level

You can export multiple connections simultaneously at the connector level. To perform the connector level migration, proceed as follows:

1. Navigate to SIMATIC S7+ Connector.
2. Select "Restore".



3. Upload the backup file.

> **NOTICE**
>
> The restore function supports two types of backups:
>
> - SIMATIC S7+ Connector backup
> - SIMATIC S7 Connector backup
>
> When using a SIMATIC S7 Connector backup, the system automatically migrates all connections for S7-1200/1500 that are configured with Optimized S7-Protocol.

This migration process ensures compatibility while maintaining all connection configurations.

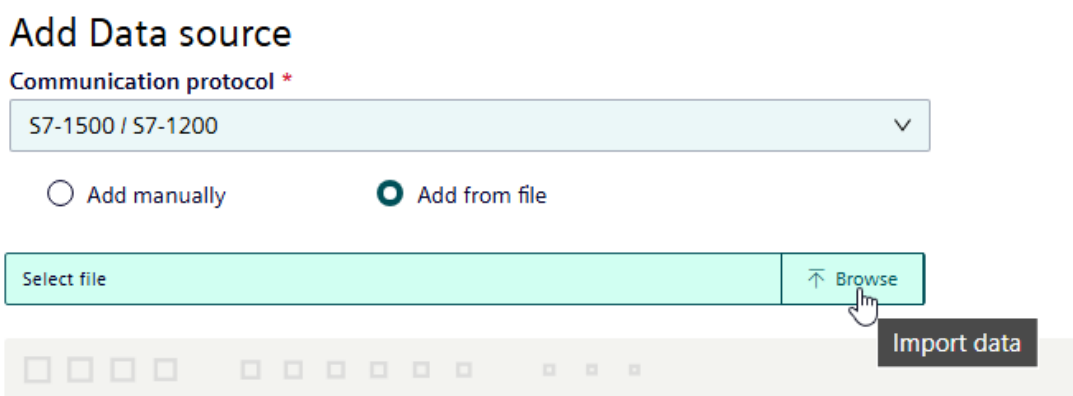
5.2.3.2 Migrating at connection level

You can migrate a single connection at the connection level. The system provides two options:

- Create a new data source from file
- Import tags for an existing data source

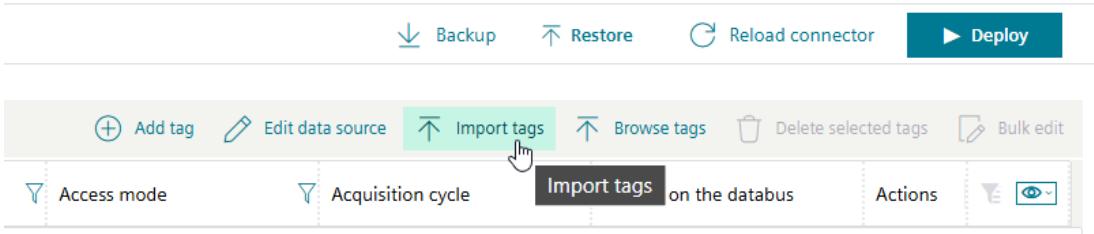
5.2.3.2.1 Creating a new data source from file

1. Select "Add Data source".
2. Choose "S7-1500 / S7-1200" from the "Communication protocol" dropdown.
3. Select "Add from file".
4. Click "Browse" to select your backup file.



5.2.3.2.2 Importing tags for an existing data source

1. Navigate to the data source where you want to import tags.
2. Click "Import tags" from the toolbar.



3. Select your backup file.
4. Follow the import wizard to complete the process.

Both options support backup files from SIMATIC S7 Connector.

> **NOTICE**
>
> When migrating an existing configuration at the connection level, ensure that the source configuration contains only one connection.

5.3 Configuring PLC alarms

5.3.1 Configuring PLC alarms

You can subscribe to PLC alarms (system events and program alarms) with the SIMATIC S7+ Connector.

SIMATIC S7-1500 CPUs provide full-text messages. The available languages depend on your PLC project.

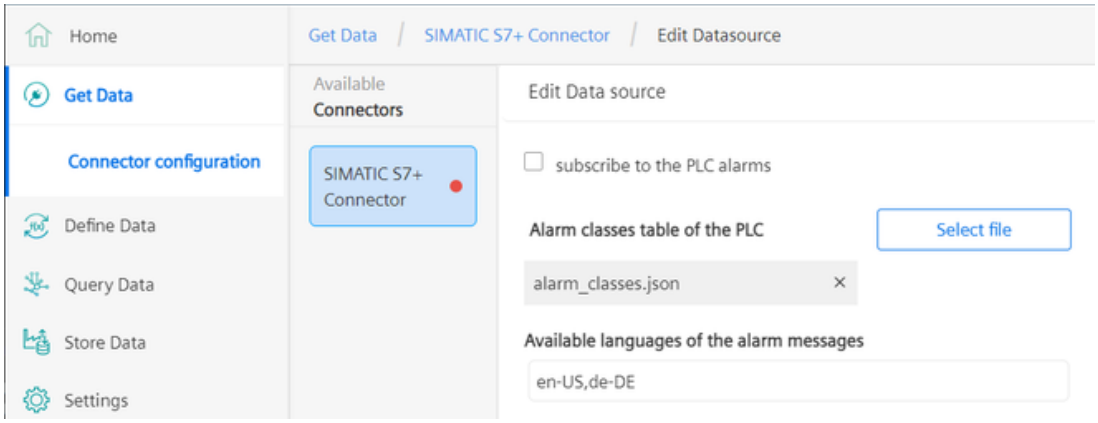
> **NOTICE**
>
> Alarm timestamps are always created from the source, i.e. the PLC. The parameter "use source timestamp" does not apply for alarms, as alarm timestamps are intrinsically tied to their generation at the PLC level.

5.3.1.1 Configuring alarm subscription

The TIA Portal export contains the alarm classes in the project languages. The Common Import Converter inserts these alarm classes into the `Config.JSON` file as an embedded file. The Common Configurator then forwards the `Config.JSON` file to the SIMATIC S7+ Connector.

To enable alarm processing, proceed as follows:

1. Open "Edit Data source".
2. Enable "Subscribe to the PLC alarms".



5.3.1.1.1 Alarm languages and Alarm classes

The connector imports the "Alarm Classes" and "Available Alarm Languages" parameters from your TIA Portal project. The connector requires this information to provide complete alarm metadata to connected clients.

Client applications can subscribe to selected alarm classes or languages. The connector configuration must reflect the full information as configured in the PLC project.

> **NOTICE**
>
> Do not change these parameters in the connector configuration. Changes affect all connected clients and make the modified data unavailable to them.

5.3.2 Support of full-text messages

Status information is provided via two APIs:

- Driver-Event-API: Status information about the CS driver, for example driver initialized, configured connections established.
- Alarm-Event-API: Interface for alarms. Full-text messages are provided via the Alarm-Event-API.

The CS client can request the alarm metadata to enable a subscription with a specific filter setting. If the client does not request the alarm metadata or does not provide a filter configuration in the subscription, it receives all alarm notifications.

The filter settings and properties for alarm notifications of the Connectivity Suite are reproduced in SIMATIC S7+ Connector.

To use the processing of full-text messages in the S7+ Connector, enable the "Subscribe to the PLC alarms" function. You can find additional information in the section: Configuring PLC alarms.

All data required for configuration are provided by the TIA Portal export and transferred directly to the Common Configurator.

5.3.3 Metadata of alarms

SIMATIC S7+ Connector receives the following metadata. You can use this metadata as filter criteria.

Metadata / Filter criterion	Functionality with SIMATIC S7+ Connector
areas	Alarm domains of the SIMATIC PLC, for example System Diagnostic, Security Event, Display Class
sources	Configured connections of SIMATIC S7+ Connector
origins	Not supported
classes	Alarm classes, as configured in the TIA project
well-known keys	A collection of possible extended alarm properties, for example: • `AlarmInfoText` • `AddlText1 … AddlText16` • `ClientAlarmID` • `ProducerID` • `ap1 .. ap16`
languages	Languages created for runtime in the TIA Portal project

> **NOTICE**
>
> The `classes` and `languages` metadata can be connection-specific for each PLC.
>
> However, the metadata and filter configuration are valid for the entire connection, which can serve multiple PLCs.

5.3.4 Properties of alarm notifications

Some of the properties of alarm notifications are specifically implemented for the SIMATIC S7+ Connector.

5.3.4.1 Alarm notifications for the SIMATIC S7+ Connector

Property	Functionality in the SIMATIC S7+ Connector
alarm ID	Unique STRING composed of PLC-specific IDs: `CRID` + `ALID` + `ARID` Example: `CTR-RID8a7e0000-ALID10d-ARID8008`
area	Alarm domain of the notification Example: `System Diagnostics`
source	Connection that triggered the alarm Example: `CPU1505S-V2.1-OC-1`
timestamps	Source generates time stamps. Example: `2022-10-28T14:44:24.1798480Z` If the PLC clock has not been synchronized, the notifications contain incorrect time stamps.
origin	Not supported
class	Alarm classes, as configured in the TIA Portal project Example: `No Acknowledgment`
alarm text	Provided in the configured languages.

5.3.4.2 Extended alarm notifications

Extended property	Functionality in SIMATIC S7+ Connector
AlarmInfoText	If available, the alarm info text is indicated. `SIMATIC Alarm Text Index == 0`
addlText n	If available, all alarm texts whose index is not equal to 1 are indicated as additional text.
ap *n*	If available, the associated parameters 1 ... n are provided as part of the extended properties.
ClientAlarmID	The alarm ID of the client is always indicated.
ProducerID	The ID of the `producer` is always specified.

5.3.4.3 Example of an alarm notification

The example shows an alarm notification that a client receives.

5.3.4.4 Alarm-Event-API notification

```
ID'CTR-RID8a7e0000-ALID10d-ARID8008'
raise-time'2022-10-28T14:44:24.1798480Z'
prio'0' area'System Diagnostic' source'CPU1505S-V2.7-OC-2' class'No Ack

is_active 'yes' is ackn. 'n.r.' is confirmed 'n.r.'

'de-DE' - 'Zustandsmeldung der CPU: CPU nicht in RUN Aktueller CPU-Betr
'en-US' - 'CPU status message: CPU not in RUN Current CPU operating moc

'AlarmInfoText' - 'de-DE' - 'Kurzbezeichnung: CPU general Bestellnummer
'AlarmInfoText' - 'en-US' - 'Short name: CPU general Order number: 6ES7

'AddlText1' - 'de-DE' - 'Software PLC_1'

'AddlText1' - 'en-US' - 'Software PLC_1'

'ClientAlarmID' - '35'

'ProducerID' - 'System Diagnostic (Innovation PLC)'

'ap0' - '01 01 00 '

'ap1' - '40 04 0c 01 00 00 00 00 10 23 00 32 00 02 00 00 ...'
```

5.4 Using special tags

5.4.1 Using special tags

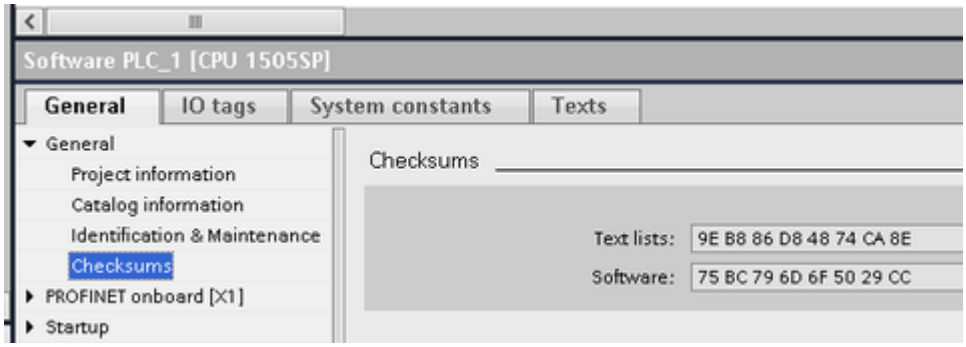
Special tags are supported in S7+ Connector 1.3 in the following manner:

- The special tags are added as valid tags.
- The JSON configuration of the S7+ Connector is extended with these tags.
- The special tags are integrated in the meta data.

Only one special tag is implemented in version 1.3:

Name	Address string	Data type
$$ChecksumStandardBlocks	8721:V:userProgCRC	String

You can check the CRC of software with this special tag.



5.4.1.1 Using special tags with IIH Essentials

If you configure special tags in IIH Essentials, no further action is required. The S7+ Connector inserts the special tags in the meta data. IIH Essentials processes the meta data and shows you the special tags.

5.4.1.2 Using special tags with the Common Configurator and Databus

The Common Configurator works with configuration files. If the configuration is generated on the basis of a TIAP export, the special tags are not present.

To include these special tags in the configuration, deploy the configuration in the S7+ Connector. The S7+ Connector updates the configuration. When the Common Configurator is once again synchronized with the S7+ Connector, the S7+ Connector gets the configuration with the special tags.

Performance measures 6

6.1 Quality codes

The Quality Code characterizes the quality of a value when it is transferred from a PLC to the S7+ Connector.

Use of the PLC time stamp in S7+ Connector affects the Quality Code:

Flags (bits 12 ... 15)		Description
Bit 12	Source quality	
Bit 13	Source time	The data time stamp was created and added by an external data source.
Bit 14	Time corrected	The data time stamp added by an external data source is corrected in the system.
Bit 15	Reserved	

You can find additional information on the topic of Quality Codes here: Quality code.

6.2 Quantity structures

6.2.1 Maximum number of tags

The maximum number of configurable tags for the SIMATIC S7+ Connector amounts to 40,000.

The total number of possible tags that can be used by all clients (IIH Semantics, Databus Gateway, etc.) for a PLC depends on the PLC type.

Maximum number of tags that can be used simultaneously for each PLC:

PLC type	Maximum number of tags
S7-1510 to S7-1513	2,000
S7-1515	4,000
S7-1516 (and all other PLCs not named here)	8,000
S7-1517	20,000
S7-1518	40,000

The maximum number of tags is a total number that is shared among other connections. If other devices (e.g. HMI) also communicate with the PLC, the number of possible subscriptions for the SIMATIC S7+ Connector is reduced. Per connection a small number of tags is used for internal purposes (currently a maximum of five internal tags).

The limits also depend on the firmware version, and the numbers here refer to firmware 2.9.

For example, if an HMI uses 150 Tags and two tags are used for internal purposes, there remain 1848 tags for a S7-1510.

If the user configures too many tags for a PLC, the PLC does not have enough memory, and the log entry **Not Enough Memory Available** appears.

The minimum cycle time is 100 ms for all PLC types.

These limits are determined by the firmware of the PLC.

NOTICE

The maximum number of tags cannot be combined with the lowest cycle time.

6.2.2 SIMATIC S7+ Connector

6.2.2.1 Read test

6.2.2.1.1 Test environment

- SIMATIC S7-1516-3 PN/DP
- SIMATIC IPC227E (IE Device)S
- SIMATIC S7+ Connector
- Databus Gateway
- Databus
- Flow Creator

6.2.2.1.2 Test scenario

Number of tags	Data type	Number of connections	Server speed	IE Device acquisition cycle	IE Device	Data loss	CPU of the IE Device	Memory in MB
40000	INT 16	20	1 s	500ms	IPC 427E	0.0055	61-69%	135
39998	INT 16	1	1 s	500ms	IPC 427E	0	50-60%	146
10000	INT 16	1	200ms	100ms	IPC 427E	0	58.83-71.90%	41
20000	INT 16	1	200ms	100ms	IPC 427E	0	87.54-99.61%	72
23000	INT 16	1	200ms	100ms	IPC 427E	0	92.46-109.44%	83
24000	INT 16	1	200ms	100ms	IPC 427E	0.0031	92.84-110.45%	83
25000	INT 16	1	200ms	100ms	IPC 427E	0.0589	90.59-101.09%	87
30000	INT 16	1	200ms	100ms	IPC 427E	0.8568	99.89-103.22%	105
10000	INT 16	1	500ms	250ms	IPC 227E	0	53.21-58.17%	42
10000	INT 16	1	1 s	500ms	IPC 227E	0	22.94-30.23%	38
13000	INT 16	1	500ms	250ms	IPC 227E	0.0088	71.91-77.61%	47
30000	INT 16	100	2000ms	1000ms	SCALANCE LPE9413	0	94%	110

6.2.2.2 Write test

6.2.2.2.1 Test environment

- SIMATIC S7+ Connector
- Databus Gateway
- Databus
- SIMATIC 1518F

- SIMATIC IPC427E

6.2.2.2.2 Test scenario

The tags are written continuously to the PLC while the subscription remains open to read value changes.

Number of tags	Data types	Server speed	IE Device acquisition cycle	Data loss
100	UINT8, UINT16, UINT32, INT8, INT16, INT32, REAL	1 s	500ms	0.001

6.2.3 Common Import Converter

6.2.3.1 Test environment

- CS Registry
- Common Configurator
- Common Import Converter
- SIMATIC S7+ Connector

6.2.3.2 Test scenario

Parameter	Details
Number of tags	150,000
Average length of the tag names	125 characters
Size of the export file	50 MB

The **Common Import Converter** converts all tags used in a TIAP project. In the next step, the user selects the tags he wants to record.

Therefore, the number of tags in the **Common Import Converter** is much larger than in the **SIMATIC S7+ Connector**.

6.2.3.3 Test result

The test was successful.

If you are processing larger export files, the configured memory for the **Common Import Converter** on the IE Device may not be sufficient. In this case, the conversion fails.

For larger export files, follow all the size and quantity restrictions listed above.

Limitations 7

7.1 Known restrictions

7.1.1 Databus Gateway

7.1.1.1 Supported JSON formats

For publishing subscribed data points, only the **JSON Timeseries** Format is supported. The variants "JSON Bulk" and "Binary" are not supported by the Databus Gateway.

7.1.1.2 Configuration update

The Databus Gateway requires restart after deploying new configurations. Communication will be temporarily interrupted during this restart. The Service will be unavailable for a brief period.

7.1.2 SIMATIC S7+ Connector

7.1.2.1 Configuration update

- System restart required after deploying new configurations
- During the restart process, there is a brief interruption in communication services

7.1.2.2 Data Type limitations

7.1.2.2.1 Structured Data Types and PLC Data Types (UDT)

- No direct support for structured data types and PLC data types (UDT)
- Individual elements must be selected separately

7.1.2.2.2 Bool Array support

- Supported: Scalar `Bool` values and 1-dimensional arrays
- Not supported: 2-dimensional arrays (must access elements individually)

7.1.2.2.3 String Array support

- Supported: Reading and writing operations
- Limitations for non-optimized data blocks:
  - Writing fails with strings configured with odd maximum string length [1)]
  - Only strings with even maximum string length are written correctly [1)]

[1] Only the configured maximum string length is relevant, not the actual string length.

Kind of data point	Example	Optimized Data Block (DB)		Non-optimized DB	
		Read	Write	Read	Write
String, even max. len	`text : String[32]`	✅	✅	✅	✅
String, odd max. len	`text : String[31]`	✅	✅	✅	✅
String-array, even max. len	`text : Array[0..9] of String[32]`	✅	✅	✅	✅
String-array, odd max. len	`text : Array[0..9] of String[31]`	✅	✅	✅	❌

NOTICE

When working (writing) with string arrays in non-optimized DBs, ensure the maximum string length is even.

7.1.2.3 PLC Timestamps

- PLC Timestamps supported only for subscriptions
- Read requests use local IE Device clock timestamps

7.1.2.4 SIMATIC S7-1500 Software Controller

- S7-1500 Software Controller accessible only via PN interface

7.1.2.5 Variable naming restrictions

- Supported: Unicode Characters up to `U+0FFFF`
- Not supported: Unicode Characters from `U+10000` and above

7.1.2.6 File size limitations

The size of the TIA Portal Scada export file is limited to 80 MB.

7.1.3 SIMATIC S7+ Connector and Databus Gateway

These restrictions apply when both apps are used together.

Function	Support by Databus Gateway
Subscribe tags	Supported [1]
Write tags	Supported [1]
Read tags	Not supported
Driver Status Events	Supported
PLC alarms	Not supported

[1] The **SIMATIC S7+ Connector** and **Databus Gateway** support all SIMATIC basic data types including `DTL`, `String` and `WString` (except the outdated types `Counter` and `Timer`). Single values and 1-dimensional arrays are supported, n-dimensional arrays are supported only for Subscription, not for Write.

7.1.4 All CS Connectors

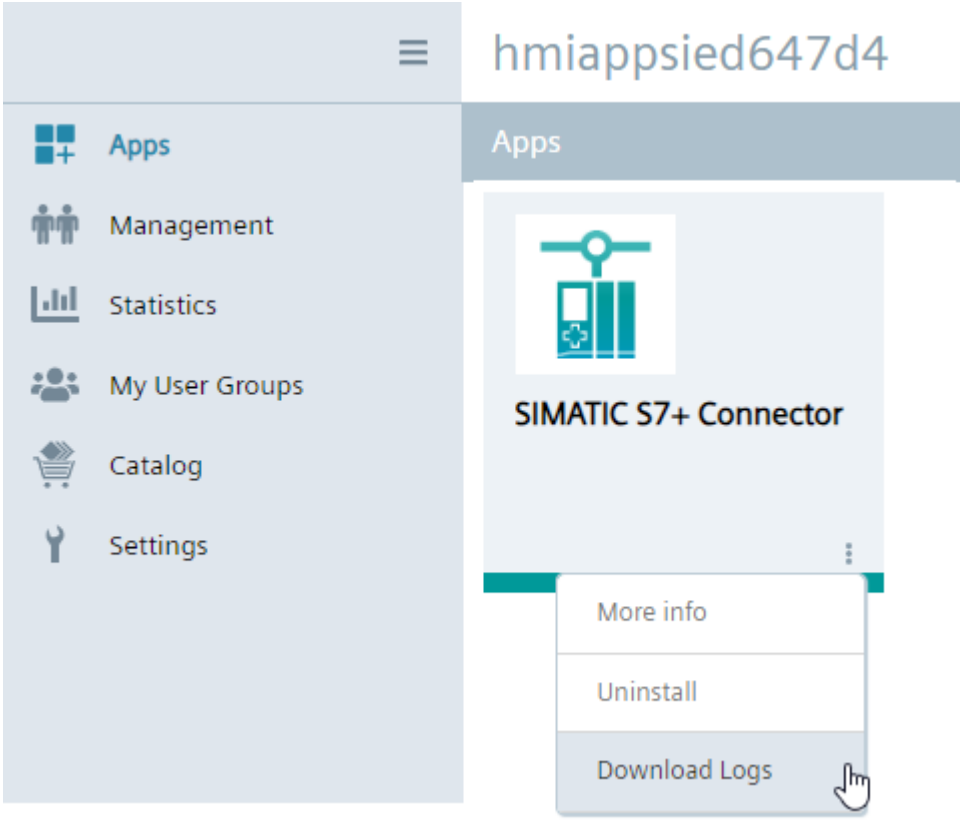
- Diagnostics are available only to a limited extent, e.g. based on the app log file.
- No support of **data records**.

7.1.5 Common Configurator

- Databus Gateway configuration: Dedicated Device data types are not supported. The Databus Gateway uses **default edge data types**.
  This restriction applies only if you create the configuration of the subscriptions with the Common Configurator. If you create the configuration manually, there are no restrictions.
- S7+ Connector Configuration: Manual addition of tags for S7+ Connector is not supported due to the missing type-safe address information. Use **Browsing** or **Import** instead.

7.1.6 System diagnosis

At the moment, the system diagnosis functionality is limited. If an Industrial Edge App does not operate as expected, the application log file can be a useful source for troubleshooting.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